



Tri-I Biotech, Inc.
源資國際生物科技股份有限公司

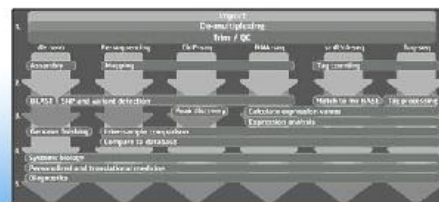
Spatial Transcriptomics Sequencing Service

源資國際生物科技股份有限公司 陳良坤

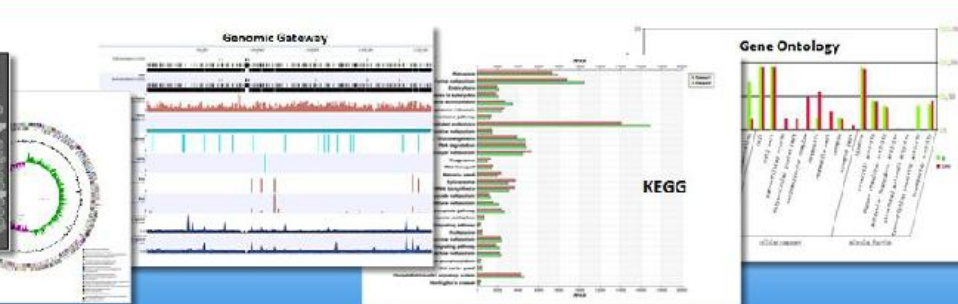




第二代定序服務 | NGS



BIOINFORMATICS



最新消息

NEWS



2012.4 2012.4 CLC 台灣區用戶研討會

2011.12 慶祝MiSeq次世代定序正式上線服務，即日起提供前五位用戶上機優惠

2011.11 本公司正式代理NuGEN產品

2011.09 本公司第二代測序技術服務開跑囉~!

2011 "2011 CLC-Bio台灣區用戶研討會"於6/17...

2011 我們搬家了!本公司自四月起遷至新北市汐止區...

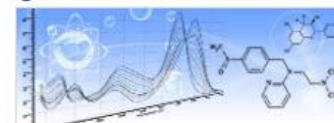
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生物資訊軟體

Bioinformatics

定序引子服務

Sequencing&Oligo



化學資訊軟體

Chemoinformatics

試劑

Reagents



Systems Biology

1. **GeneGO (USA)**- Integrate pathway analysis database
2. **Softgenetics (USA)**-Genetic and SNP analysis tools
3. **Funakoshi (Japan)**—molecular biology related reagents

Chemoinformatics

1. **Certara (USA)**- drug discovery
2. **ACD Labs (Canada)**- Chemical/ NMR analysis solution...
3. **Gaussian (USA)**-predict electronic structure programs

Bioinformatics

1. **Genious (USA)**-DNA analysis software
2. **Bioinformatics solution (Canada)**- Structure biology solution
3. **Gene Codes (USA)**-DNA sequencing data analysis
3. **Prism (USA)**-Statistical analysis and drawing

Service @ Tri-I Biotech :

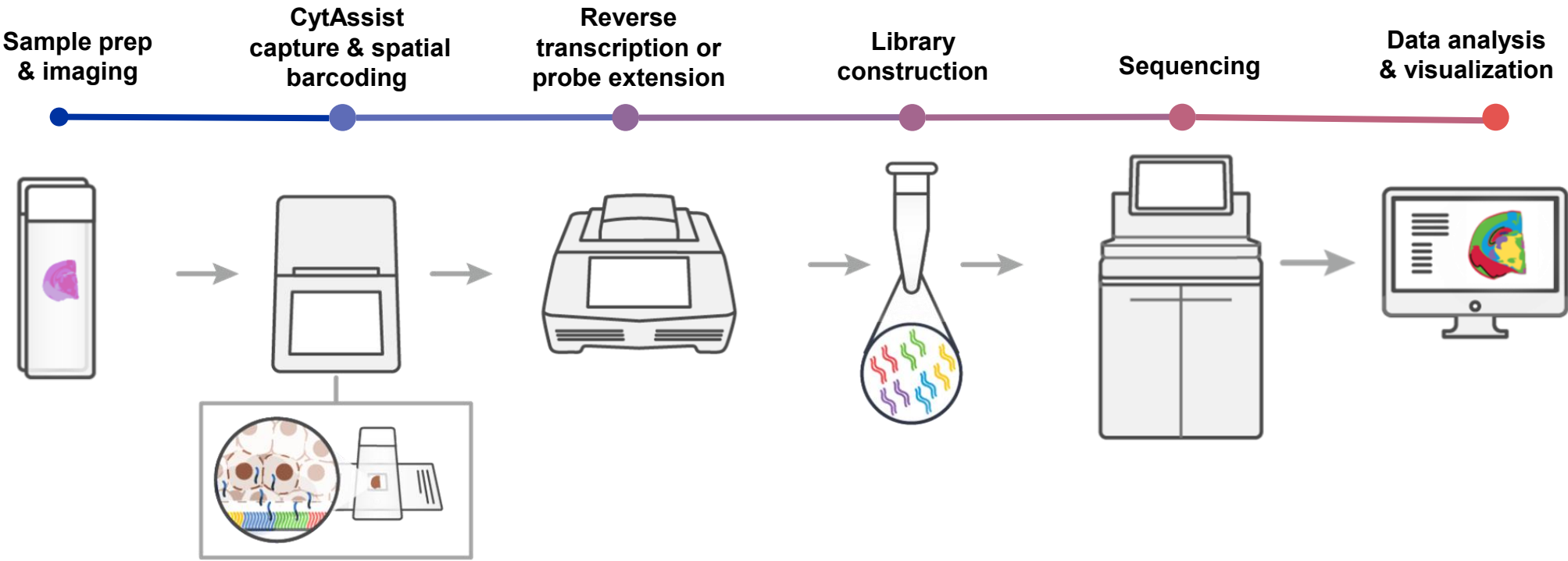
1. Sanger Sequencing
2. NGS Sequencing
3. Oligo synthesis
4. Genotyping & SNP analysis
5. Gene Expression analysis (qPCR & RNA-seq)
6. Bioinformatics analysis



- **Spatial transcriptomics – Visium system**

Transform histological sections into whole transcriptome data

End-to-end solution for single cell-scale spatial transcriptomics



Spatial Transcriptomics: A Powerful Tool in Disease Understanding and Drug Discovery

Junxian Cao^{1,4}, Caifeng Li¹, Zhao Cui², Shiwen Deng^{1,4}, Tong Lei³, Wei Liu^{1✉}, Hongjun Yang^{1,2,4✉}, Peng Chen^{1,4✉}

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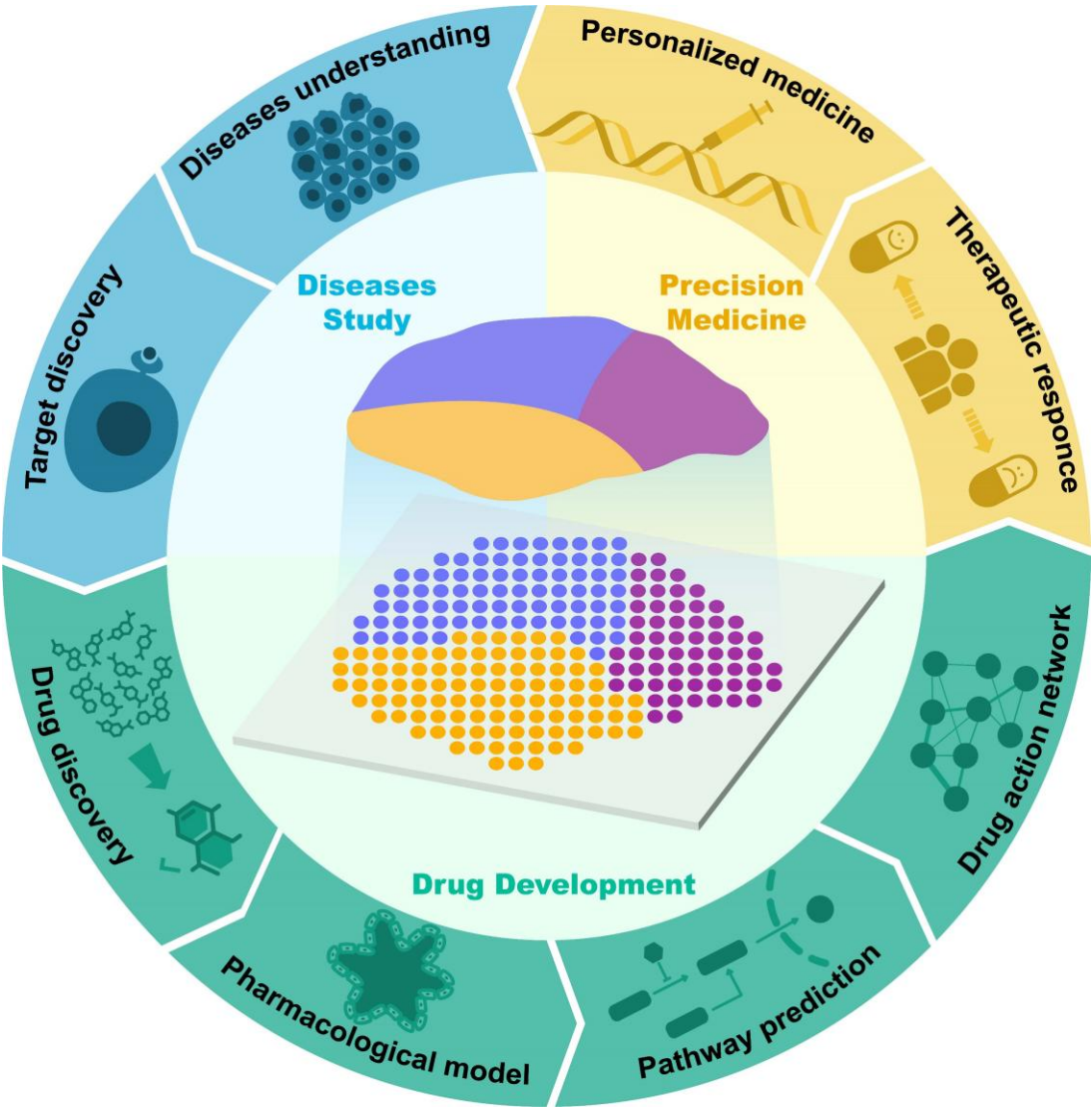
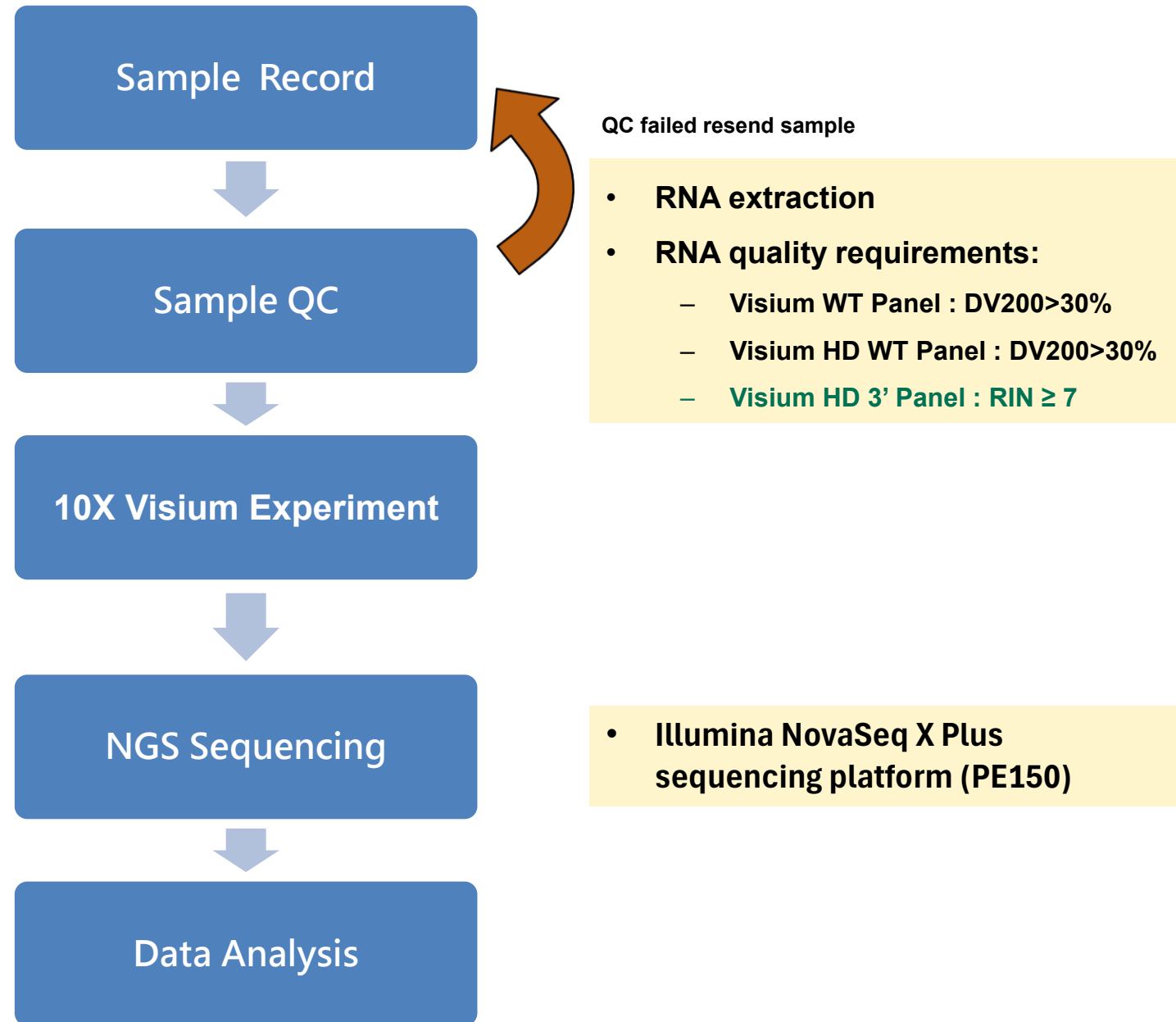


Figure 1. Application of spatial transcriptomics in drug research and development.

Three types of Visium

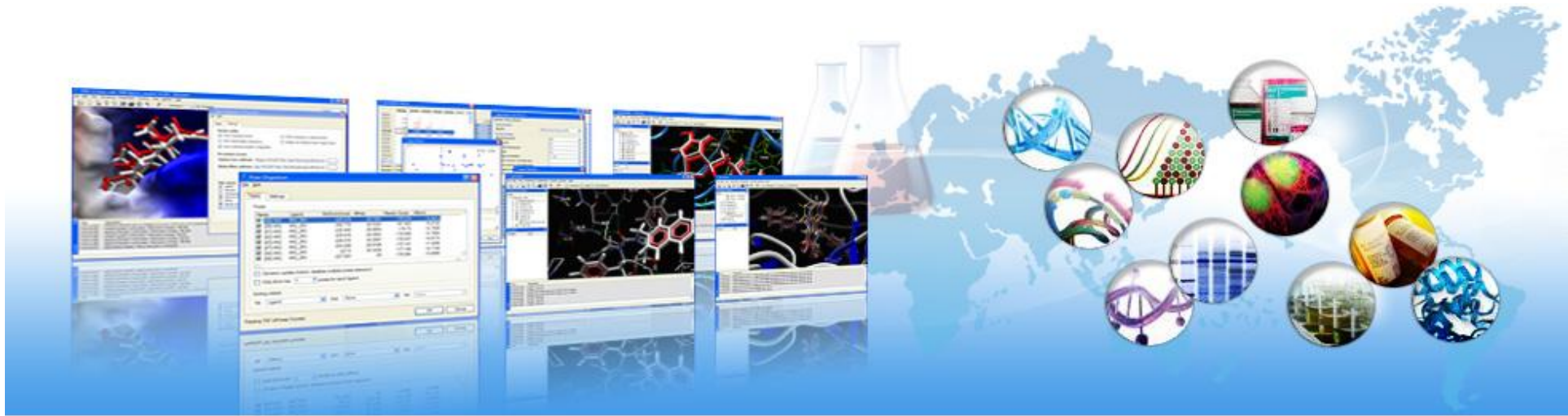
	Visium WT Panel Gene Expression	Visium HD WT Panel Gene Expression	Visium HD <u>3'</u> Gene Expression
Resolution	55um per spot (1-10 cell level)	2um per bin (single cell level)	2um per bin (single cell level)
Applicable species	Whole Transcriptome Probe	Whole Transcriptome Probe	Poly-A mRNA capture
	Human & Mouse	Human & Mouse	All Species agnostic 3' whole transcriptome
Applicable samples	• FFPE • fresh-frozen • fixed-frozen tissues	• FFPE • fresh-frozen • fixed-frozen tissues	Only fresh frozen tissues
Acceptable specimen types	• FFPE wax block • FFPE H&E stained tissue slide • FFPE IF stained tissue slide	• FFPE wax block • FFPE H&E stained tissue slide • FFPE IF stained tissue slide	OCT H&E stained tissue slide

Workflow



Demo report

for Single cell RNA-seq service



◆ Space Ranger analysis_ Summary report

10X
GENOMICS

Space Ranger • count

Visium_HD_Human_Colon_Cancer - Gene expression library of Colon Cancer (Visium HD) using the Human Whole Transcriptome Probe Set

Summary Image Alignment Bin-Level Metrics Cell Segme

Key Metrics

516,880

Number of 8 µm binned squares under tissue

1395.3

Mean reads per 8 µm bin

523.0

Mean UMIs per 8 µm bin

18,118

Total genes detected

Mapping ?

Reads Mapped to Probe Set	98.8%
Reads Mapped Confidently to Probe Set	98.2%
Reads Mapped Confidently to the Filtered Probe Set	96.3%
Reads Half-Mapped to Probe Set	0.5%
Reads Split-Mapped to Probe Set	0.0%

Sequencing ?

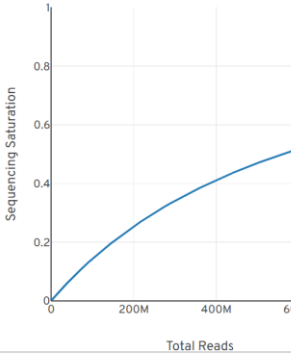
Number of Reads	721,219,716
Valid Barcodes	90.0%
Valid UMI Sequences	99.9%
Sequencing Saturation	56.5%
Q30 Bases in Barcode	94.8%
Q30 Bases in Probe Read	94.1%
Q30 Bases in UMI	95.2%
Fraction of Bins Under Tissue 8 µm	73.6%
UMIs per sq mm of Tissue	8,172,085
Reads per sq mm of Tissue	19,457,827
Fraction Reads in Squares Under Tissue	98.9%

Cell Segmentation Metrics ?

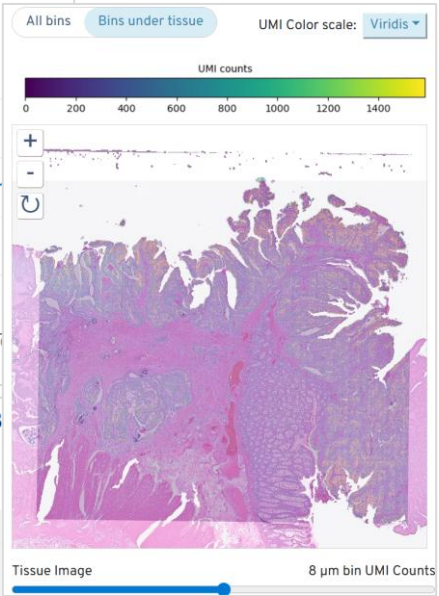
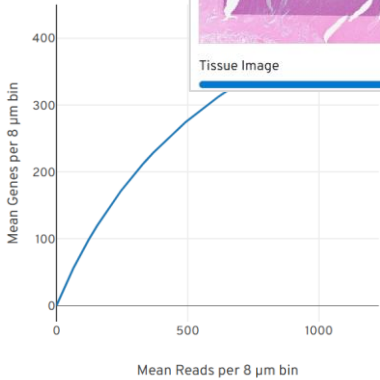
Number of Cells	220,703
Reads in Cells	80.0%
UMIs in Cells	87.9%
Mean Reads per Cell	2,586.2
Median Genes per Cell	580.0
Median UMIs per Cell	759.0
Median Cell Area (µm ²)	96.0
Median Nucleus Area (µm ²)	28.0
Maximum Nucleus Diameter (pixels)	256

Sequencing Depth Evaluation

Sequencing Saturation ?

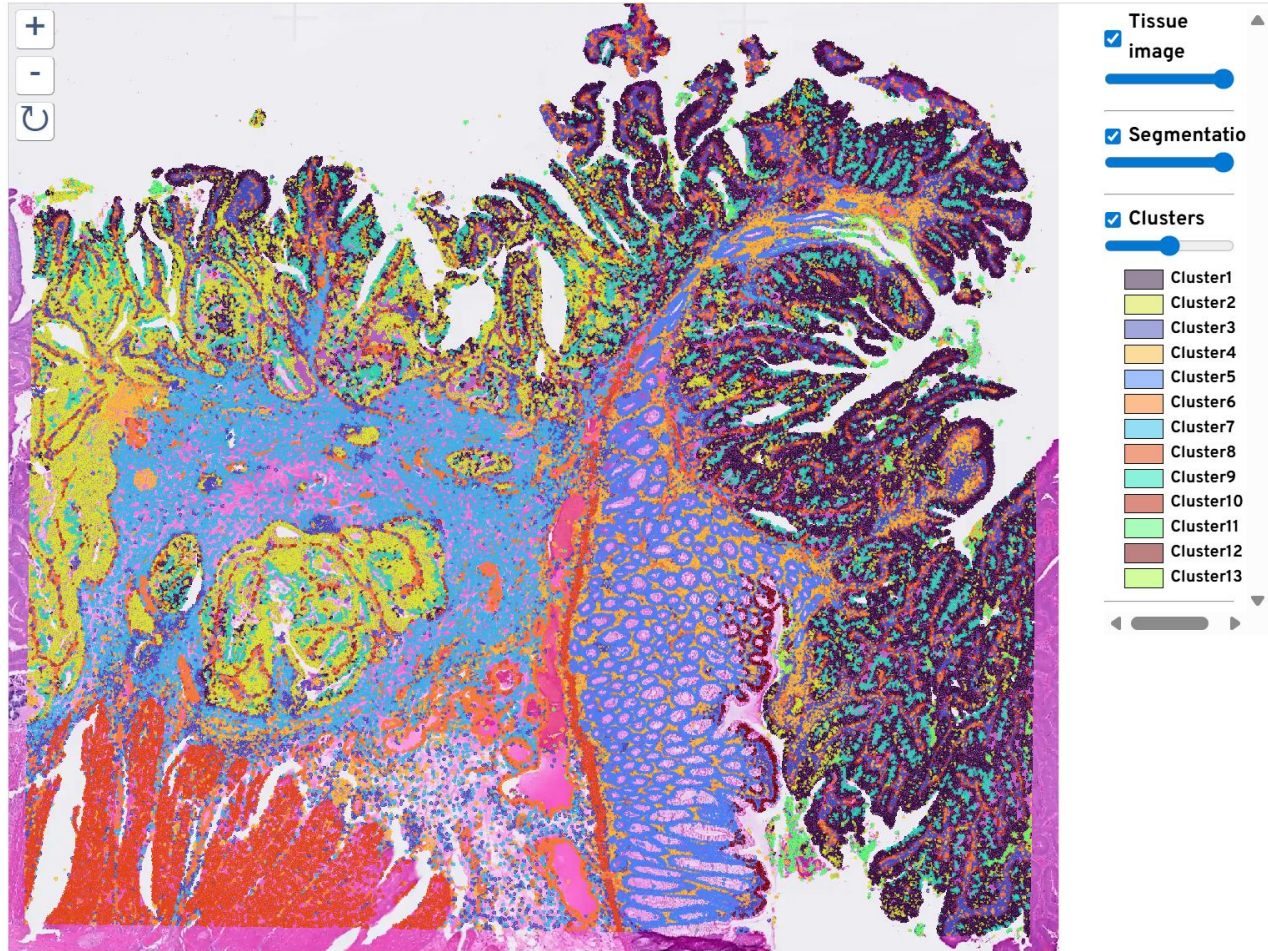


Median Genes per 8 µm bin



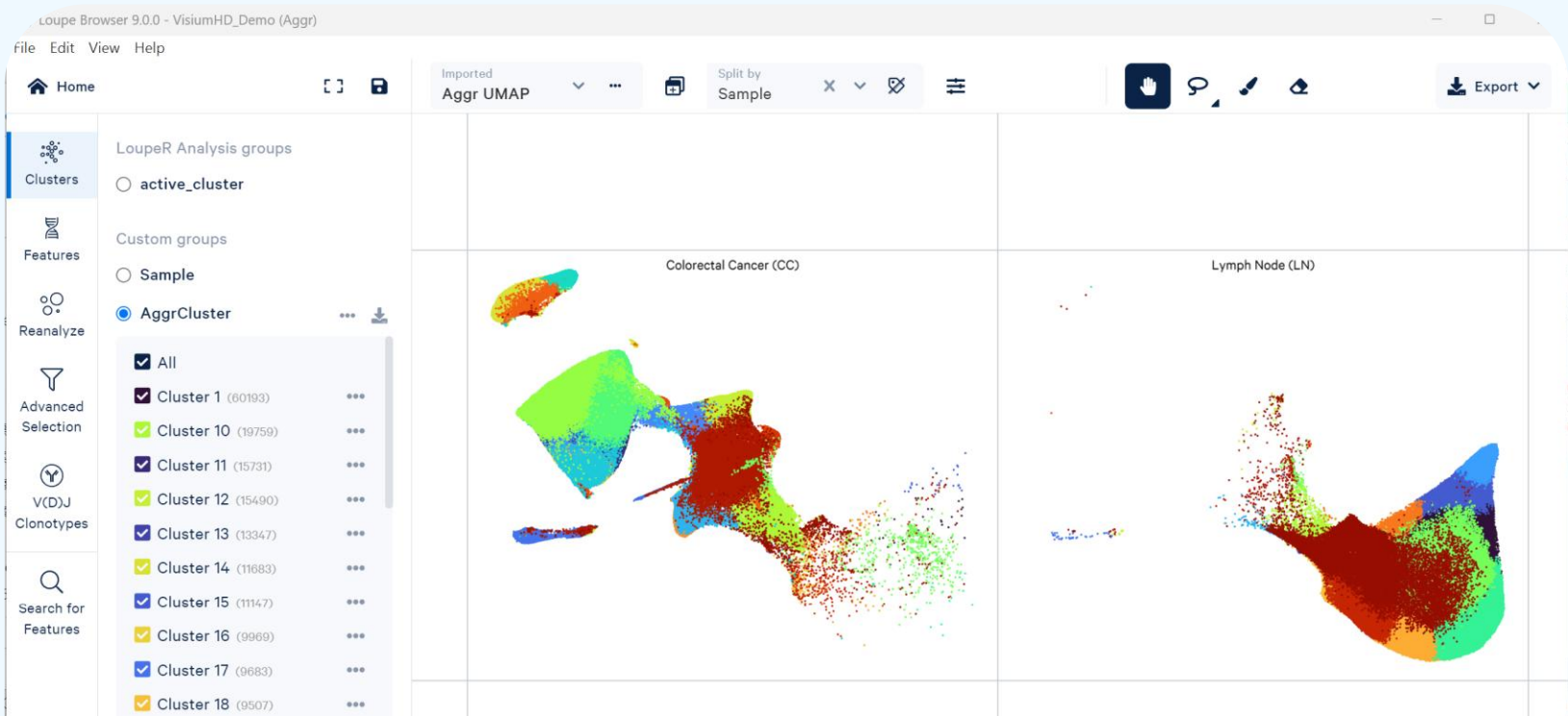
◆ Space Ranger analysis_ Summary report

Cell Segmentation ?



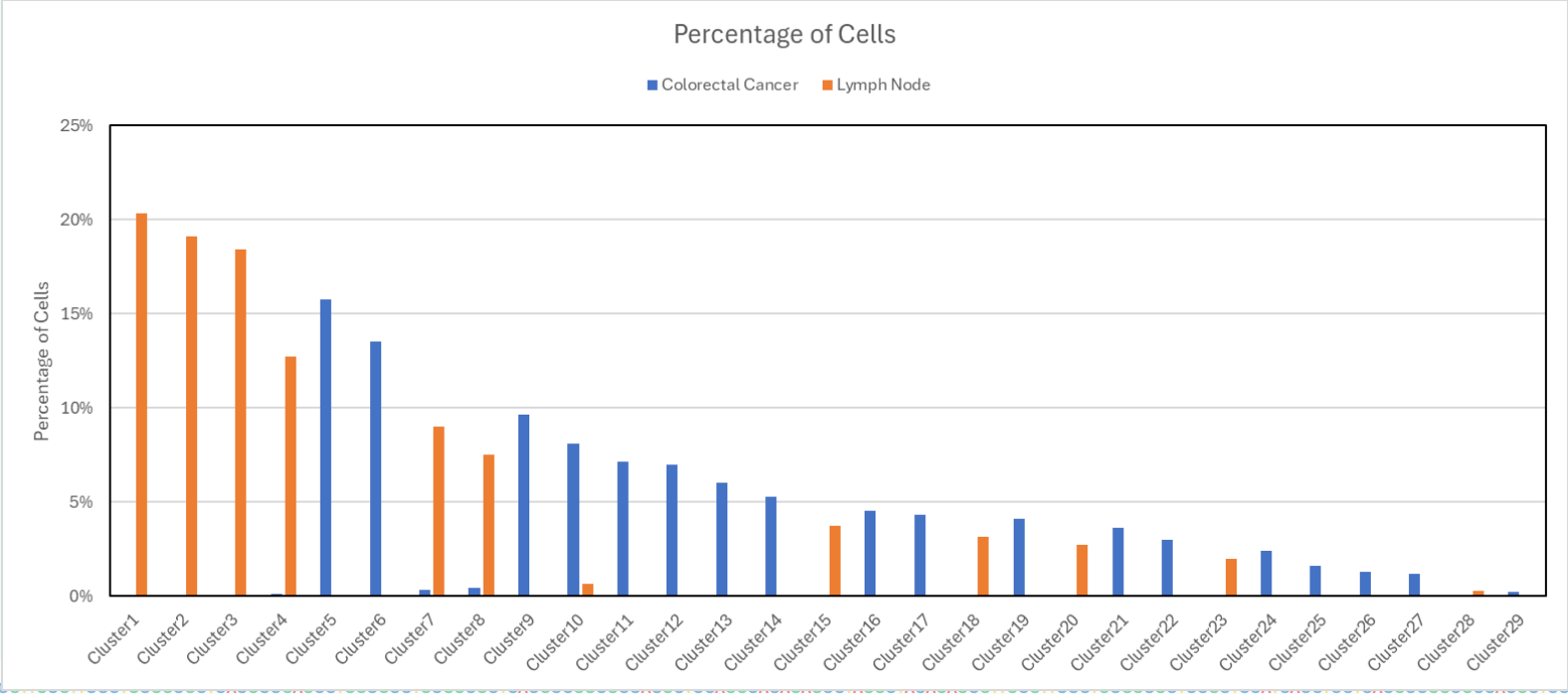
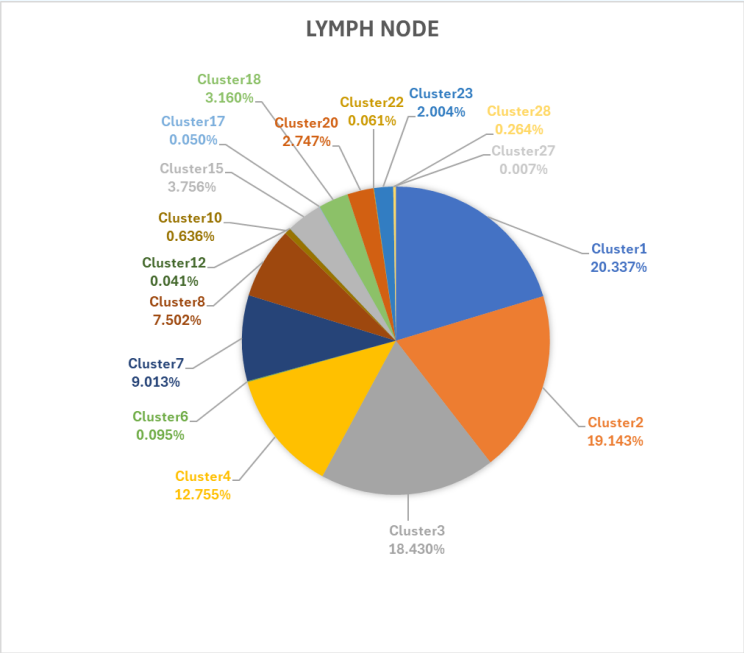
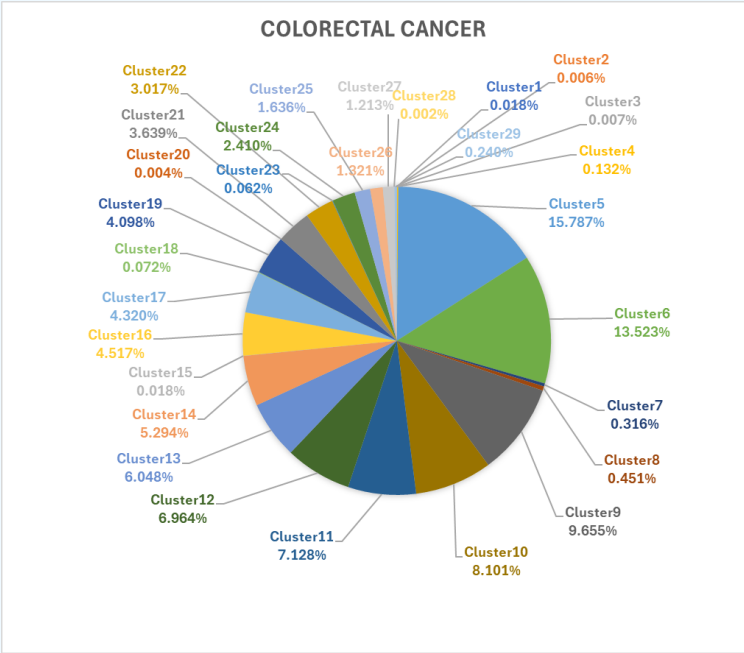
◆ Aggregation

Loupe Browser <https://www.10xgenomics.com/support/software/loupe-browser/latest>



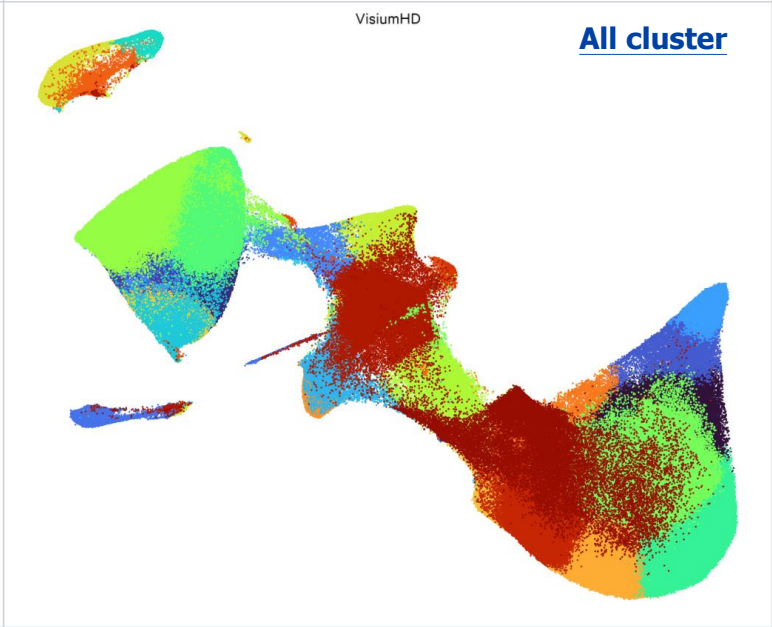
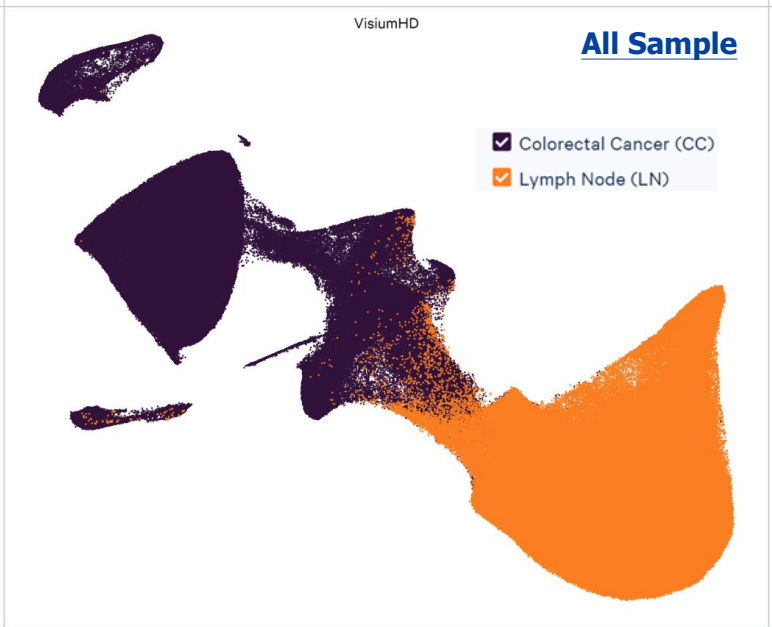
Sample	Total Reads (single end)	Mapping rate (Reads mapped to probe set)	Uniquely Mapped Rate	Cell number (Segmentation)	Median genes per cell	Median UMI counts per cell	Total genes detected
Colorectal Cancer	721,219,716	98.8%	98.2%	220,703	581	759	18,132
Lymph Node	443,996,591	98.9%	96.3%	295,776	68	72	18,132

	Cell Count		Percentage of Cells	
Cluster	Colorectal Cancer	Lymph Node	Colorectal Cancer	Lymph Node
Cluster1	40	60153	0.018%	20.337%
Cluster2	13	56619	0.006%	19.143%
Cluster3	16	54511	0.007%	18.430%
Cluster4	291	37725	0.132%	12.755%
Cluster5	34841	0	15.787%	0.000%
Cluster6	29845	281	13.523%	0.095%
Cluster7	698	26657	0.316%	9.013%
Cluster8	996	22188	0.451%	7.502%
Cluster9	21307	0	9.655%	0.000%
Cluster10	17879	1880	8.101%	0.636%
Cluster11	15731	0	7.128%	0.000%
Cluster12	15370	120	6.964%	0.041%
Cluster13	13347	0	6.048%	0.000%
Cluster14	11683	0	5.294%	0.000%
Cluster15	39	11108	0.018%	3.756%
Cluster16	9969	0	4.517%	0.000%
Cluster17	9534	149	4.320%	0.050%
Cluster18	160	9347	0.072%	3.160%
Cluster19	9044	0	4.098%	0.000%
Cluster20	9	8126	0.004%	2.747%
Cluster21	8030	1	3.639%	0.000%
Cluster22	6659	179	3.017%	0.061%
Cluster23	136	5928	0.062%	2.004%
Cluster24	5319	1	2.410%	0.000%
Cluster25	3611	0	1.636%	0.000%
Cluster26	2916	0	1.321%	0.000%
Cluster27	2676	21	1.213%	0.007%
Cluster28	5	782	0.002%	0.264%
Cluster29	530	0	0.240%	0.000%
Total	220694	295776	100.000%	100.000%

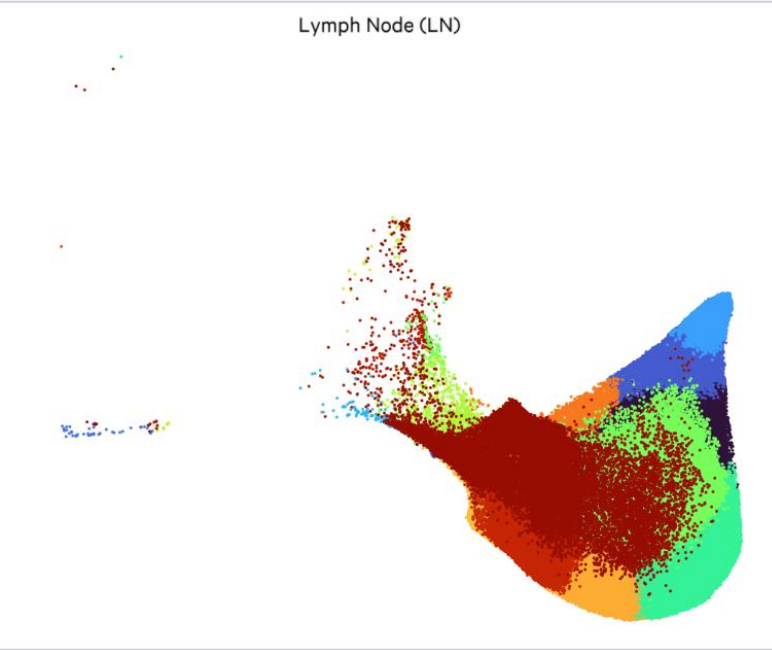
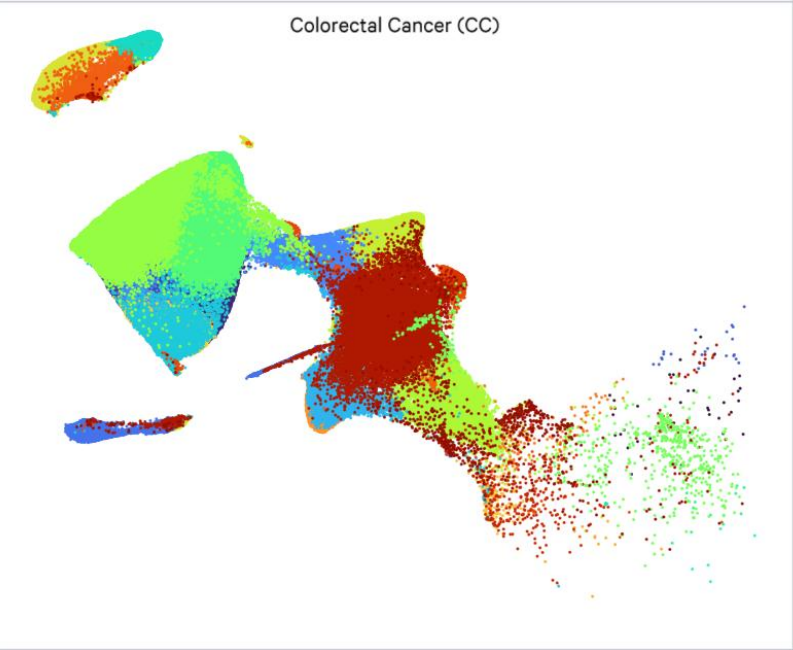


◆ Cluster Analysis

UMAP圖

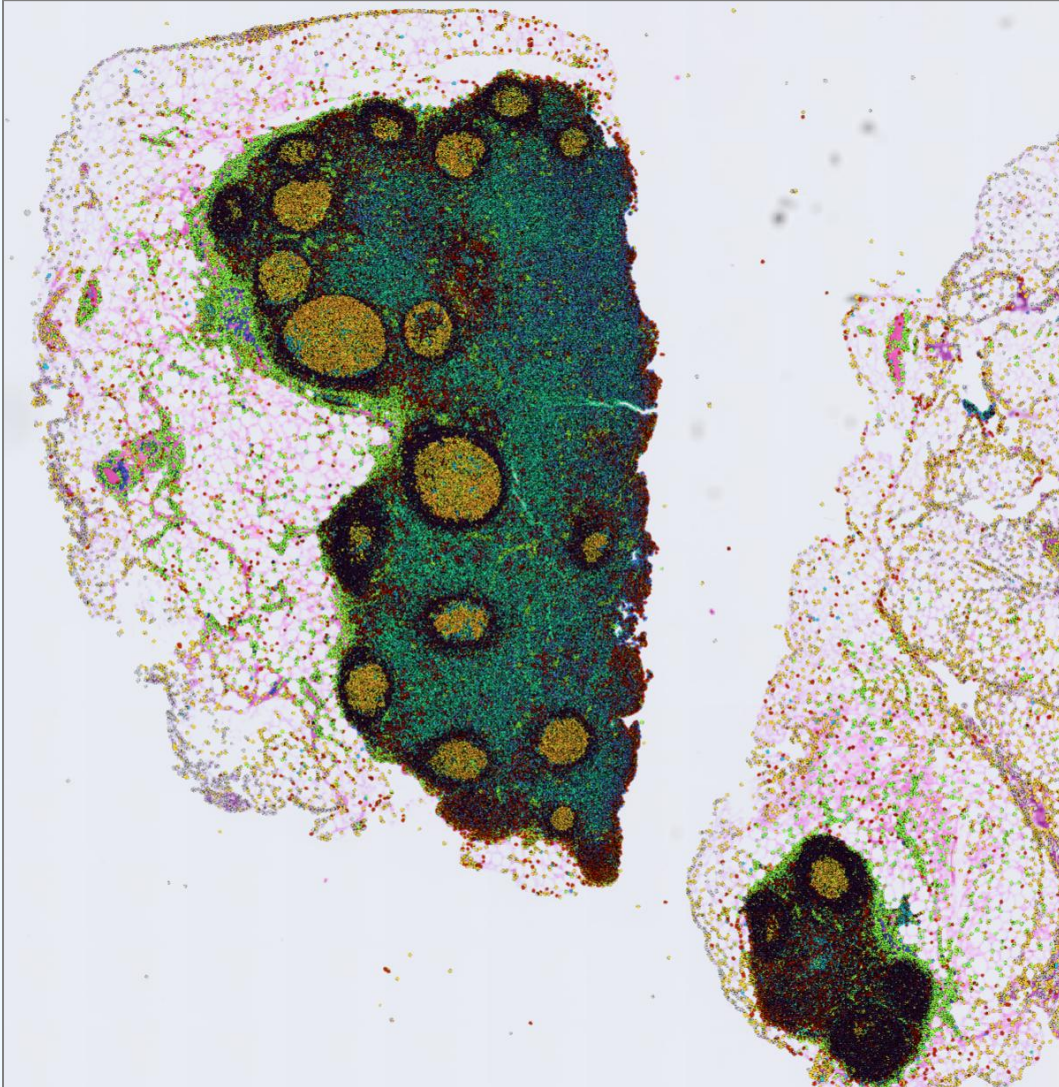


Per Sample
Per Cluster

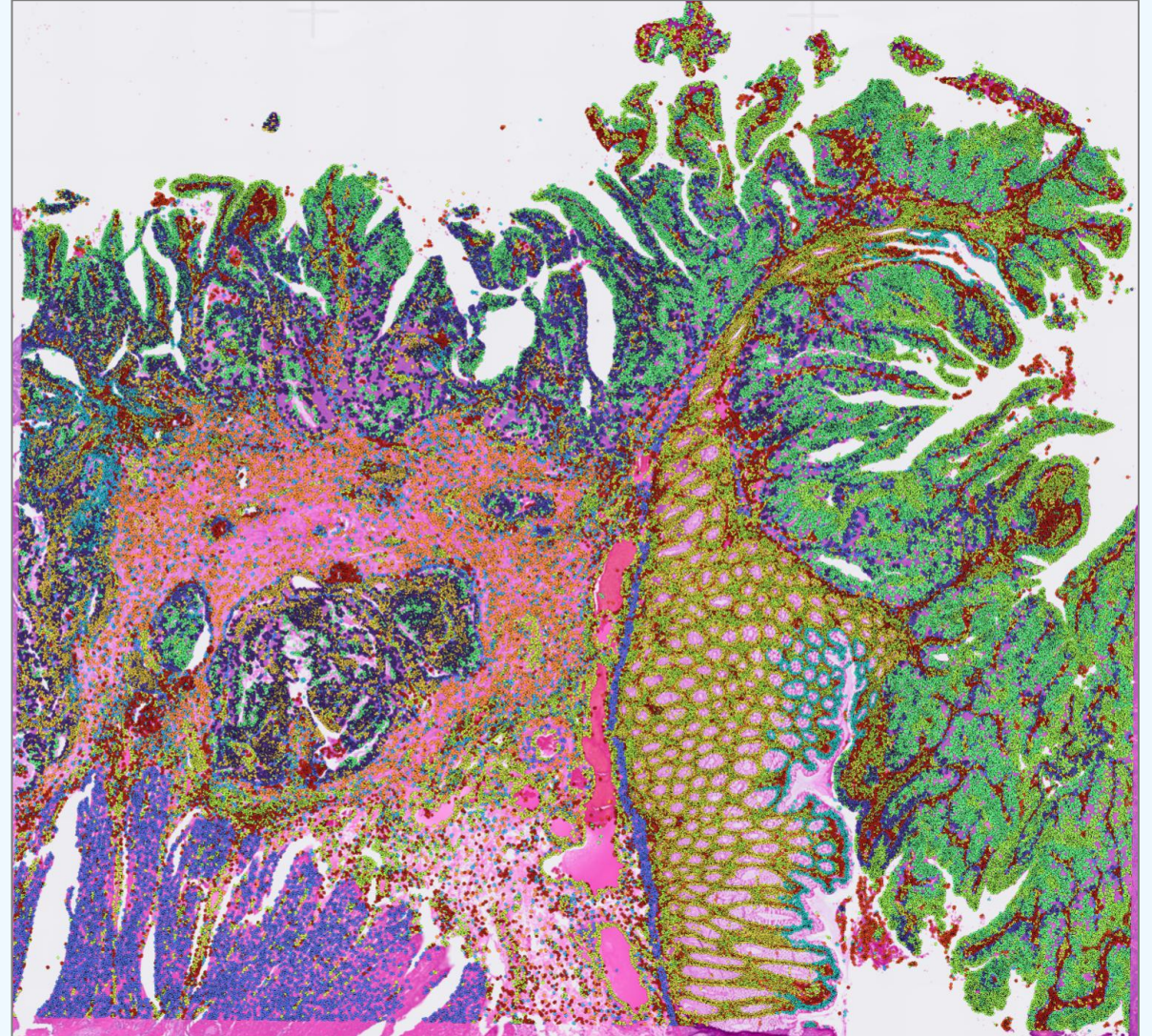


◆ Aggregation Cluster

Lymph Node



Colorectal Cancer



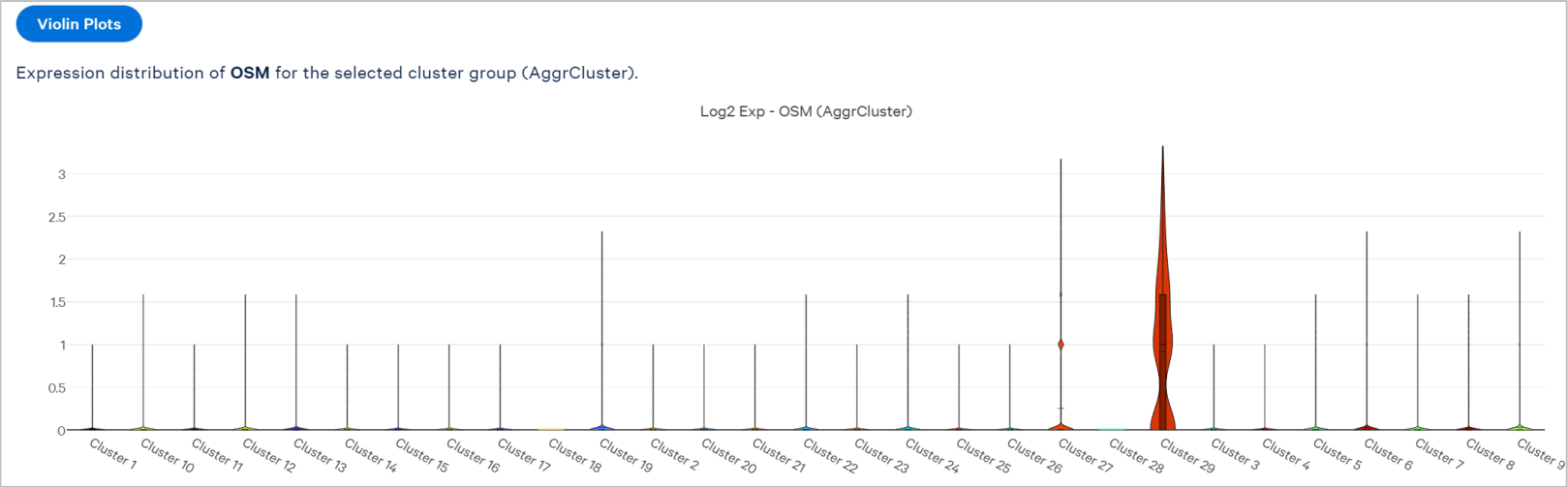
◆ Top50 _ all significant genes per cluster

FeatureID	FeatureName	Cluster 1 Average	Cluster 1 Log2 Fold Change	Cluster 1 P-Value	Cluster 2 Average	Cluster 2 Log2 Fold Change	Cluster 2 P-Value	Cluster 3 Average	Cluster 3 Log2 Fold Change	Cluster 3 P-Value	Cluster 4 Average	Cluster 4 Log2 Fold Change	Cluster 4 P-Value	Cluster 5 Average	Cluster 5 Log2 Fold Change	Cluster 5 P-Value	Cluster 6 Average	Cluster 6 Log2 Fold Change	Cluster 6 P-Value	Cluster 7 Average	Cluster 7 Log2 Fold Change
ENSG00000089199	CHGB	79.841426	0.6032189	5.37E-07	70.944307	0.345631	0.2923813	67.430458	0.4169437	3.46E-06	61.355318	0.14624	0.280403	87.154649	0.643739	0.0781895	47.118417	-0.265791	0.625864	63.184419	0.1612442
ENSG00000160182	TFF1	6.0124344	0.5984503	7.24E-07	5.0290536	0.2500594	0.5574927	5.7145305	0.7158977	4.98E-16	3.4506909	-0.356983	0.0059337	5.9530185	0.4946664	0.2548615	2.7679717	-0.633741	0.1284037	4.5046605	0.0786855
ENSG00000145321	GC	15.836504	0.5688425	2.68E-06	13.148499	0.2139228	0.6604332	14.848249	0.6353395	6.79E-13	10.805613	-0.092935	0.5476889	17.329216	0.618922	0.0978028	7.0202566	-0.713276	0.0788368	13.383977	0.2290381
ENSG00000166509	CLEC3A	2.1850824	0.5682813	3.37E-06	1.9336543	0.3101064	0.390912	1.9959954	0.5662626	2.61E-10	1.5150149	-0.06602	0.6887205	2.2687169	0.540503	0.1898581	1.394838	-0.178472	0.7957105	1.8568995	0.2379994
ENSG00000165646	SLC18A2	1.9839112	0.5391253	1.10E-05	1.9424175	0.4390063	0.121529	1.8194546	0.5345806	2.33E-09	1.4937542	0.0481948	0.7585265	1.701065	0.2326453	0.8243025	1.808107	0.3234813	0.4131917	1.4430514	-0.010665
ENSG00000173702	MUC13	1.3199764	0.5302964	1.68E-05	1.1239381	0.2197613	0.6476287	1.2654123	0.6370965	8.88E-13	0.9874517	0.0284086	0.8614835	1.2690635	0.3949898	0.4621464	0.7640824	-0.353107	0.4731341	0.9945354	0.0341704
ENSG00000173467	AGR3	1.8910093	0.5146983	2.97E-05	1.6972993	0.2861978	0.4545917	1.8325841	0.6426541	4.77E-13	1.3061216	-0.123763	0.4042129	1.8239538	0.3866701	0.4776992	1.0838897	-0.38065	0.4250885	1.5037203	0.0986233
ENSG00000156234	CXCL13	8.2905585	0.5033234	4.39E-05	7.0736245	0.1997492	0.7053427	8.359559	0.7321977	9.68E-17	4.912644	-0.392176	0.002332	8.8286171	0.523911	0.2097235	3.8009226	-0.717612	0.0775109	6.9249135	0.1590725
ENSG00000106541	AGR2	3.6611063	0.4975545	5.58E-05	3.5743686	0.3991367	0.1828801	3.4566524	0.5493898	7.31E-10	2.7232178	-0.013243	0.9516507	3.6271111	0.4119374	0.4192855	2.7732871	0.0152046	1	3.0854433	0.1678563
ENSG00000100604	CHGA	295.4563	0.4939181	5.96E-05	286.59721	0.3861179	0.2047029	248.48062	0.2589569	0.0046932	282.09425	0.4256116	0.0003051	315.83138	0.5216422	0.2107453	257.20211	0.2175827	0.6215254	266.75651	0.263821
ENSG00000198886	MT-ND4	57.351534	0.482606	8.99E-05	64.22822	0.5956751	0.0142186	49.135601	0.2878988	0.0015665	46.727761	0.1305736	0.3422443	50.475641	0.2239858	0.8356859	56.110671	0.3813632	0.3075305	47.997711	0.1446118
ENSG00000198727	MT-CYB	40.453223	0.4822041	9.16E-05	44.7079	0.5750525	0.0196396	34.82685	0.2989727	0.0010057	33.519637	0.159789	0.2308044	35.102249	0.2027035	0.8747467	41.588675	0.454703	0.2012608	31.625794	0.0455481
ENSG00000124107	SLPI	1.1561756	0.4871805	9.31E-05	0.993993	0.1977629	0.7139825	1.1269017	0.6169296	5.25E-12	0.7125008	-0.342013	0.0091312	1.1570456	0.4185848	0.4122158	0.6148095	-0.512939	0.2447962	0.8818648	0.0172388
ENSG00000135346	CGA	8.5214863	0.4763893	0.0001421	7.3698193	0.1973075	0.7147995	6.7422818	0.0946131	0.333267	9.8037878	0.7562561	1.39E-11	10.055615	0.6541905	0.0757984	6.4116483	-0.01245	1	7.1134202	0.1361828
ENSG00000198938	MT-CO3	112.50737	0.4687436	0.0001498	121.17275	0.52397	0.0407991	100.76247	0.3762231	2.92E-05	90.160304	0.0878951	0.54534	101.54446	0.2496683	0.7943805	109.55923	0.3627956	0.3383087	91.061745	0.0847711
ENSG00000160180	TFF3	15.718246	0.4635576	0.000184	14.17479	0.249598	0.5562221	15.76174	0.6569338	9.71E-14	11.863913	-0.021059	0.9162389	14.964551	0.3243915	0.6206848	9.3979926	-0.361404	0.453659	13.145635	0.128204
ENSG00000134240	HMGCS2	1.7253224	0.4599077	0.0002268	1.5635985	0.2541037	0.5458039	1.7195685	0.6352075	8.54E-13	1.2812677	-0.052544	0.7585872	1.829153	0.4812006	0.2796528	0.874819	-0.605488	0.1509272	1.4842196	0.1679399
ENSG00000116678	LEPR	1.397581	0.4603002	0.0002294	1.2466224	0.2304615	0.6165145	1.4907926	0.8208003	1.16E-20	0.8851012	-0.321772	0.0142127	1.272372	0.2558978	0.7871176	0.7149154	-0.592097	0.162912	1.1072061	0.0495322
ENSG00000198712	MT-CO2	50.97762	0.4479902	0.0003122	55.660601	0.5269631	0.0391443	42.242104	0.1637018	0.0802394	46.526414	0.2971732	0.0153732	43.333437	0.1437467	0.9521259	60.236716	0.6334254	0.0506995	38.784716	-0.021878
ENSG00000198763	MT-ND2	17.058689	0.4468175	0.0003266	18.710603	0.5329039	0.0361788	14.648166	0.2453112	0.0074957	14.44172	0.1624153	0.2221463	14.240925	0.1161054	0.9912945	18.411951	0.4973994	0.1522794	12.863958	-0.035535
ENSG00000002933	TMEM176A	4.4582471	0.4468397	0.0003339	3.9652057	0.2144223	0.6591195	4.4685861	0.6306639	1.01E-12	3.3857112	-0.026784	0.8883114	4.5846986	0.4247877	0.3886543	2.3493874	-0.560277	0.1888994	3.9716415	0.2066165
ENSG00000171951	SCG2	11.71473	0.4455341	0.0003494	10.541583	0.2309165	0.6102768	10.669276	0.3851675	1.92E-05	11.123085	0.3658967	0.002285	12.715685	0.5038383	0.2393521	6.8909163	-0.400296	0.3904469	10.419871	0.2028122
ENSG00000198840	MT-ND3	21.950642	0.4400054	0.0004091	24.002551	0.522319	0.0418054	19.340378	0.2973267	0.0010758	18.775836	0.173795	0.1875833	19.394221	0.1941698	0.8899605	23.8376	0.500773	0.1488947	17.498624	0.0393297
ENSG00000175426	PCSK1	13.597078	0.4399467	0.0004161	12.460716	0.2537269	0.5420237	13.203156	0.5379666	1.45E-09	11.929252	0.218611	0.0873034	13.601903	0.3779493	0.4913779	9.1424125	-0.208494	0.731238	12.003761	0.1873037
ENSG00000258818	RNASE4	2.2103685	0.4390393	0.0004458	1.9907401	0.2267798	0.6239072	2.1062663	0.4886204	4.76E-08	1.9415451	0.2198024	0.086378	2.1831673	0.3585347	0.5420885	1.403254	-0.293347	0.577126	1.8048977	0.074872
ENSG00000212907	MT-ND4L	36.657374	0.4358855	0.0004675	40.981554	0.552597	0.027199	33.431218	0.3751007	3.08E-05	29.333448	0.0529812	0.7311786	31.889179	0.1677652	0.9177847	33.864577	0.2570649	0.5410976	30.873934	0.1155609
ENSG00000118271	TTR	27.318909	0.4334852	0.000516	24.657535	0.2251492	0.6253923	24.837762	0.3642299	5.43E-05	27.505931	0.460325	8.20E-05	29.625188	0.4922639	0.2564672	17.503911	-0.284821	0.5894048	24.850386	0.2252677



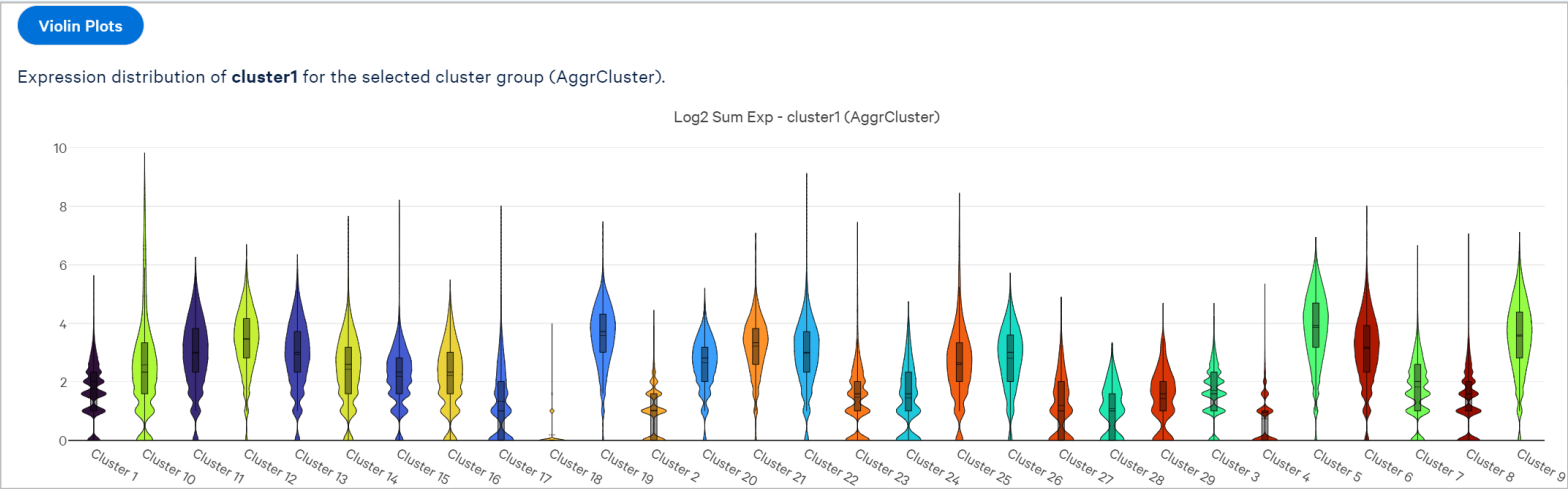
◆ Selected features (UMAP)

OSM



◆ Selected features

Name	ID
TMSB4X	feature_17428
OSM	feature_17127
IGHM	feature_12369
VIM	feature_8729



◆ Cell annotation



CellMarker 2.0

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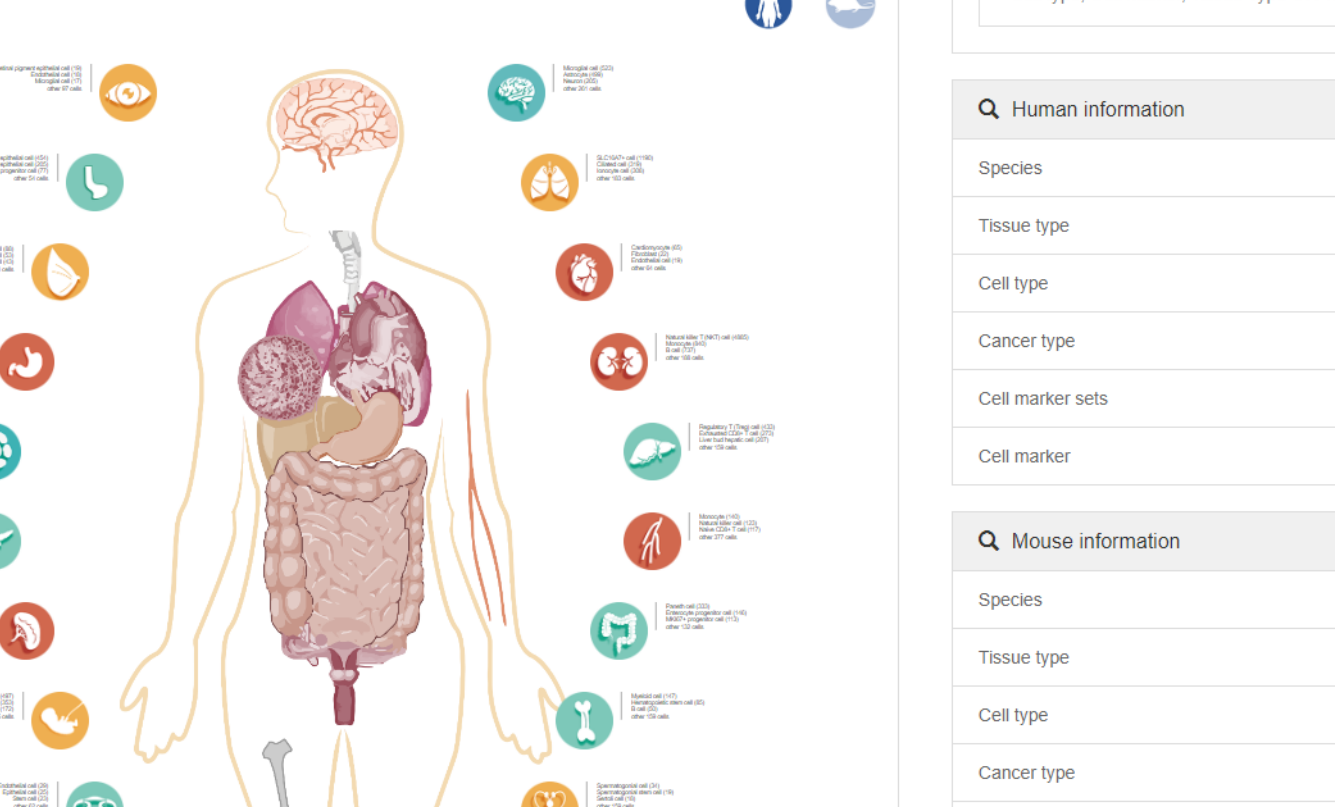
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Human

Mouse



Retinal pigment epithelial cell (15)

Endothelial cell (17)

Mitochondrion (17)

other 17 cells

Glial cell (15)

Epithelial cell (15)

Secondary progenitor cell (17)

other 15 cells

Cancer cell (15)

Cancer stem cell (15)

Epithelial cell (15)

other 15 cells

Pituitary progenitor cell (15)

Neuroendocrine progenitor cell (17)

Stem cell (15)

other 15 cells

Adipocyte-derived stem cell (15)

Adipocyte-derived stem cell (15)

Epithelial cell (15)

other 15 cells

Bone cells (15)

Neuroendocrine stem cell (15)

Adipocyte (15)

other 15 cells

Neuroblastoma cell (15)

Neuroblastoma cell (15)

Epithelial cell (15)

other 15 cells

Microglial cell (15)

Neuroblastoma cell (15)

Neuroblastoma cell (15)

other 15 cells

Epithelial cell (15)

Epithelial cell (15)

Epithelial cell (15)

other 15 cells

Neuroblastoma cell (15)

Neuroblastoma cell (15)

Neuroblastoma cell (15)

other 15 cells

Microglial cell (15)

Neuroblastoma cell (15)

Neuroblastoma cell (15)

other 15 cells

Stem cell (15)

Cancer cell (15)

Epithelial cell (15)

other 15 cells

Cardiomyocyte (15)

Endothelial cell (15)

Epithelial cell (15)

other 15 cells

Neuroblastoma cell (15)

Neuroblastoma cell (15)

Neuroblastoma cell (15)

other 15 cells

Regulatory T cell (15)

Regulatory T cell (15)

Regulatory T cell (15)

other 15 cells

Microglial cell (15)

Neuroblastoma cell (15)

Neuroblastoma cell (15)

other 15 cells

Prostate cell (15)

Epithelial cell (15)

Epithelial cell (15)

other 15 cells

Microglial cell (15)

Neuroblastoma cell (15)

Neuroblastoma cell (15)

other 15 cells

Spermatogonial cell (15)

Spermatogonial cell (15)

Spermatogonial cell (15)

other 15 cells

Cancer cell (15)

Cancer cell (15)

Cancer cell (15)

other 15 cells

Cell type, Cell marker, Tissue Type OR Tissue Cl:

Q

Q Human information

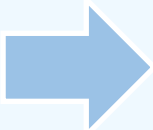
Species	Human
Tissue type	429
Cell type	1715
Cancer type	278
Cell marker sets	4334
Cell marker	15737

Q Mouse information

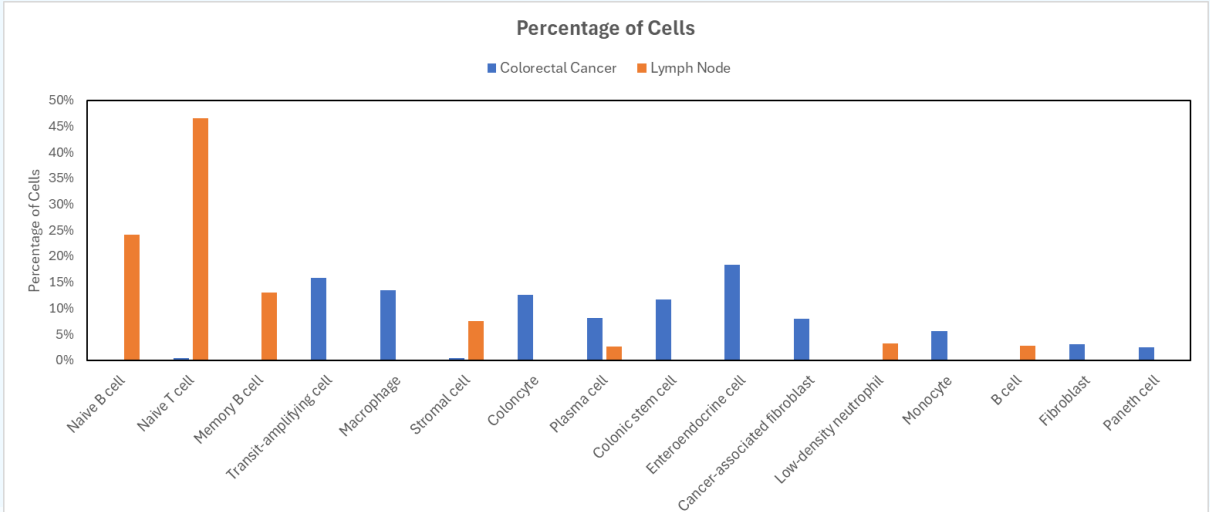
Species	Mouse
Tissue type	399
Cell type	1434
Cancer type	94
Cell marker sets	2185
Cell marker	11385

<http://bio-bigdata.hrbmu.edu.cn/CellMarker/index.html>

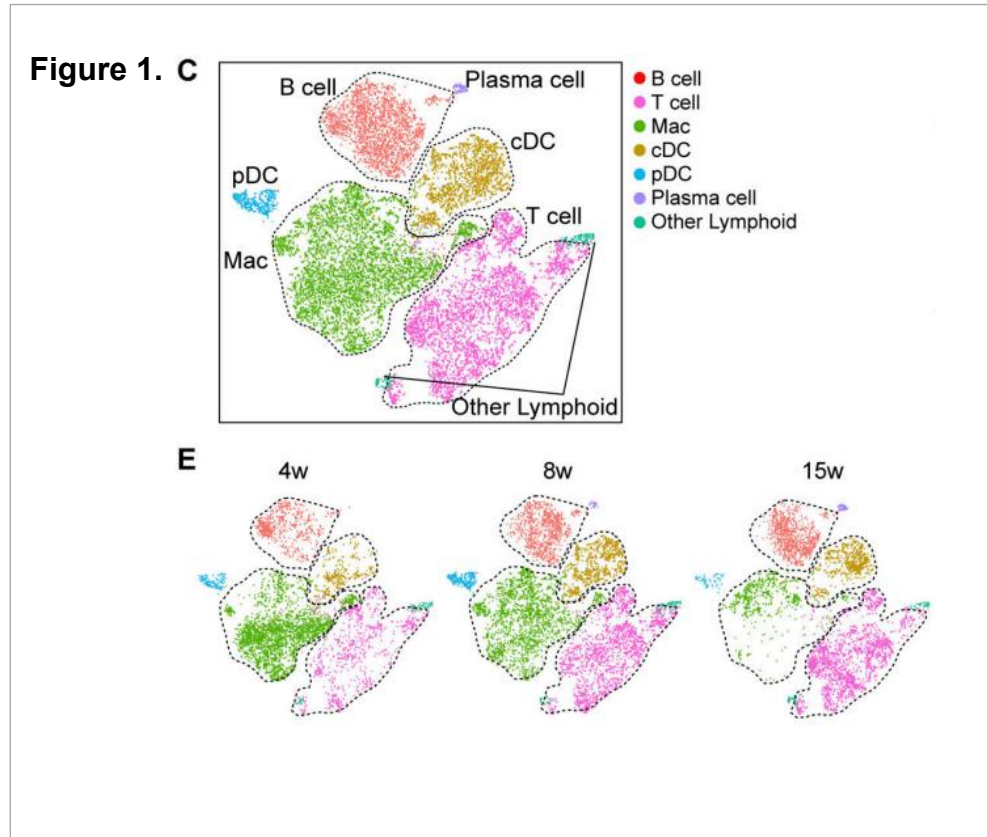
		Cell Count		Percentage of Cells	
Cluster	Cell Type	Colorectal Cancer	Lymph Node	Colorectal Cancer	Lymph Node
Cluster1	Naive B cell	40	60153	0.018%	20.337%
Cluster2	Naive T cell	13	56619	0.006%	19.143%
Cluster3	Naive T cell	16	54511	0.007%	18.430%
Cluster4	Memory B cell	291	37725	0.132%	12.755%
Cluster5	Transit-amplifying cell	34841	0	15.787%	0.000%
Cluster6	Macrophage	29845	281	13.523%	0.095%
Cluster7	Naive T cell	698	26657	0.316%	9.013%
Cluster8	Stromal cell	996	22188	0.451%	7.502%
Cluster9	Colonocyte	21307	0	9.655%	0.000%
Cluster10	Plasma cell	17879	1880	8.101%	0.636%
Cluster11	Colonic stem cell	15731	0	7.128%	0.000%
Cluster12	Enteroendocrine cell	15370	120	6.964%	0.041%
Cluster13	Enteroendocrine cell	13347	0	6.048%	0.000%
Cluster14	Enteroendocrine cell	11683	0	5.294%	0.000%
Cluster15	Naive B cell	39	11108	0.018%	3.756%
Cluster16	Colonic stem cell	9969	0	4.517%	0.000%
Cluster17	Cancer-associated fibroblast	9534	149	4.320%	0.050%
Cluster18	Low-density neutrophil	160	9347	0.072%	3.160%
Cluster19	Monocyte	9044	0	4.098%	0.000%
Cluster20	B cell	9	8126	0.004%	2.747%
Cluster21	Cancer-associated fibroblast	8030	1	3.639%	0.000%
Cluster22	Fibroblast	6659	179	3.017%	0.061%
Cluster23	Plasma cell	136	5928	0.062%	2.004%
Cluster24	Paneth cell	5319	1	2.410%	0.000%
Cluster25	Colonocyte	3611	0	1.636%	0.000%
Cluster26	Colonocyte	2916	0	1.321%	0.000%
Cluster27	Monocyte	2676	21	1.213%	0.007%
Cluster28	Memory B cell	5	782	0.002%	0.264%
Cluster29	Monocyte	530	0	0.240%	0.000%
Total		220694	295776	100.000%	100.000%



		Cell Count		Percentage of Cells	
Cell Type		Colorectal Cancer	Lymph Node	Colorectal Cancer	Lymph Node
Naive B cell		79	71261	0.036%	24.093%
Naive T cell		727	137787	0.329%	46.585%
Memory B cell		296	38507	0.134%	13.019%
Transit-amplifying cell		34841	0	15.787%	0.000%
Macrophage		29845	281	13.523%	0.095%
Stromal cell		996	22188	0.451%	7.502%
Colonocyte		27834	0	12.612%	0.000%
Plasma cell		18015	7808	8.163%	2.640%
Colonic stem cell		25700	0	11.645%	0.000%
Enteroendocrine cell		40400	120	18.306%	0.041%
Cancer-associated fibroblast		17564	150	7.959%	0.051%
Low-density neutrophil		160	9347	0.072%	3.160%
Monocyte		12250	21	5.551%	0.007%
B cell		9	8126	0.004%	2.747%
Fibroblast		6659	179	3.017%	0.061%
Paneth cell		5319	1	2.410%	0.000%
Total		220694	295776	100.000%	100.000%



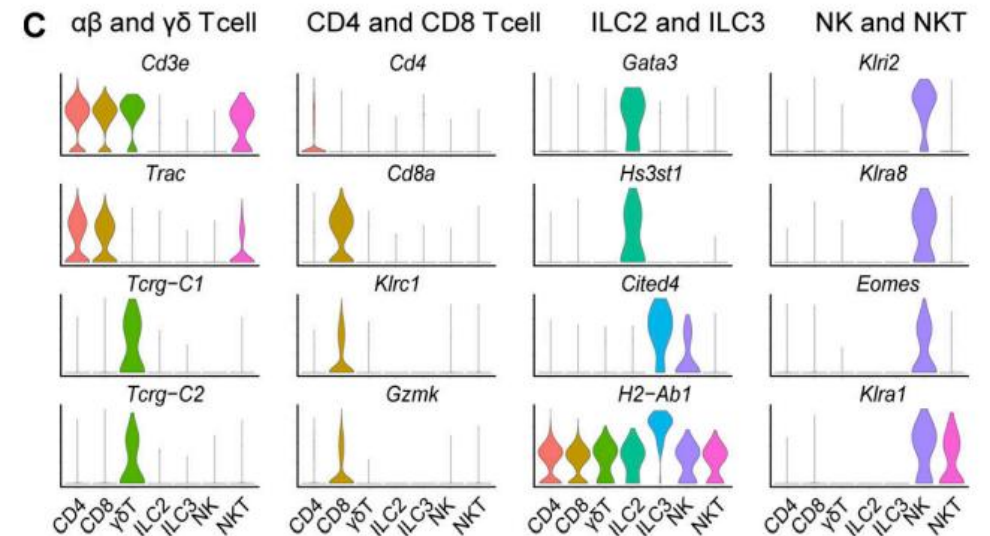
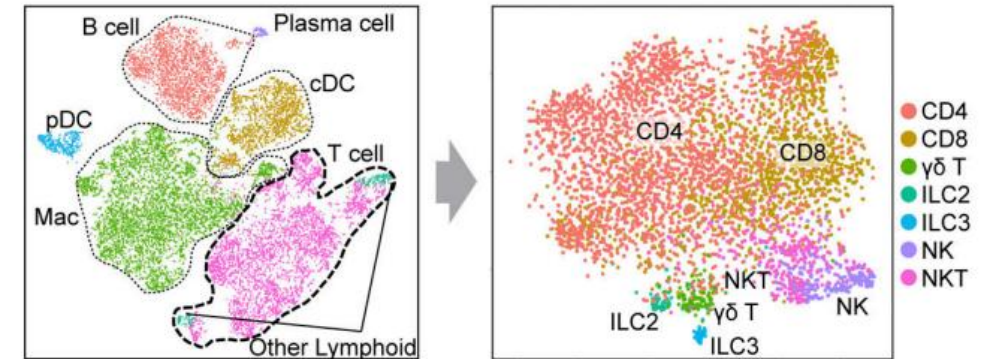
◆ Changes in cell types



Zakharov PN et al., J Exp Med. 2020 Jun 1;217(6):e20192362.

◆ Subtype clustering analysis

Figure 2. A



◆ Comparison of Gene expression

Name	Chromosome	Region	Exons	Gene ID	Gene length	Exon length	Biotype	EntrezGene ID	Gene Description	Transcripts annotated	GO Biological Process	GO Cellular Component	GO Molecular Function	KEGG Pathway	Target-based of Drugs	Pvalue - Treat vs Control	Pvalue - Treat vs Control	Log2Fold Change - Treat / Control	Log2Fold Change - Treat / Control	Mean - Cluster3 Control (0 =>)	Mean - Cluster3 Treat (0 =>)	Mean - Cluster4 Control (0 =>)	Mean - Cluster4 Treat (0 =>)	Mean - Cluster3 Control	Mean - Cluster3 Treat	Mean - Cluster4 Control	Mean - Cluster4 Treat
SCNN1D	1	1280436...	33	ENSG000	11594	4025	protein_c	6339	sodium ch	8	GO:00509	GO:00160	GO:00055	-	-	2.65E-27	1.59E-26	-0.13784	-0.06226	0.632398913	0.574774356	0.16359713	0.156686816	0.632398913	0.574774356	0.16359713	0.156686816
ACAP3	1	compleme	50	ENSG000	17220	8707	protein_c	116983	ArfGAP v	13	GO:00435	-	GO:00468	hsa04144	-	1.99E-80	2.5E-112	-0.18821	-0.18546	5.128694935	4.501418937	0.989908716	0.870494506	5.128694935	4.501418937	0.989908716	0.870494506
PUSL1	1	1308597...	19	ENSG000	3081	2459	protein_c	126789	pseudouri	5	GO:00080	GO:00432	GO:00099	-	-	1	1	-0.04578	0.049901	0.222990492	0.216025393	0.122904179	0.127229676	0.222990492	0.216025393	0.122904179	0.127229676
INTS11	1	compleme	117	ENSG000	13107	8220	protein_c	54973	integrator	50	GO:00427	GO:00057	GO:00167	-	-	7.98E-73	2.25E-18	-0.11145	0.080915	3.233003653	2.992643616	1.495631792	1.581913227	3.233003653	2.992643616	1.495631792	1.581913227
CPTP	1	1324756...	7	ENSG000	4141	2555	protein_c	80772	ceramide-	3	-	-	-	-	-	1	1	0.071789	-0.6185	0.460412511	0.483902406	0.320754873	0.208922423	0.460412511	0.483902406	0.320754873	0.208922423
TAS1R3	1	1331280...	6	ENSG000	4035	3475	protein_c	83756	taste 1 rec	1	GO:00015	GO:00058	GO:00049	hsa04973	D02381 A	1	1	-0.30558	0.561205	0.053711658	0.043458978	0.010425133	0.015382297	0.053711658	0.043458978	0.010425133	0.015382297
DVL1	1	compleme	34	ENSG000	14143	3882	protein_c	1855	dishevelle	8	GO:00219	GO:00301	GO:00198	hsa04310	-	1	2.88E-32	0.194902	-0.1065	0.854972279	0.978641054	0.467346079	0.434089417	0.854972279	0.978641054	0.467346079	0.434089417
MXRA8	1	compleme	29	ENSG000	9089	3622	protein_c	54587	matrix re	10	GO:00608	GO:00160	GO:00036	-	-	1.9E-107	5.68E-26	0.476629	1.383983	0.575018655	0.800131617	0.050450353	0.131669324	0.575018655	0.800131617	0.050450353	0.131669324
AURKA1	1	compleme	13	ENSG000	1766	1669	protein_c	54998	aurora kin	6	GO:00458	GO:00056	GO:00055	-	-	1	1.2E-24	-0.02442	-0.07434	1.18057645	1.160760492	0.780144882	0.740961991	1.18057645	1.160760492	0.780144882	0.740961991
CCNL2	1	compleme	59	ENSG000	13625	6857	protein_c	81669	cyclin L2	20	GO:00459	GO:00056	GO:00055	-	-	0	5.62E-07	0.100088	-0.10609	12.07235241	12.93961619	5.837167852	5.423332642	12.07235241	12.93961619	5.837167852	5.423332642
MRPL20	1	compleme	14	ENSG000	5385	2767	protein_c	55052	mitochond	6	GO:00701	GO:00057	GO:00198	hsa03010	-	1	1	0	0	0.000001	0.000001	0.000001	0.000001	0	0	0	0
ANKRD6	1	compleme	11	ENSG000	3350	2644	protein_c	441869	ankyrin re	5	-	-	-	-	-	1	2.26E-37	-0.25084	0.171741	0.182470507	0.1533495	0.029081626	0.032758034	0.182470507	0.1533495	0.029081626	0.032758034
TMEM88	1	1426128...	2	ENSG000	1660	492	protein_c	643965	transmem	1	-	GO:00160	-	-	-	3.99E-53	1	1.516594	2.168338	0.008699235	0.024889806	0.005717528	0.025700593	0.008699235	0.024889806	0.005717528	0.025700593
VWA1	1	1434861...	10	ENSG000	8022	5065	protein_c	64856	von Wille	4	GO:00482	GO:00700	GO:00428	-	-	3.67E-32	9.62E-60	0.289349	1.178082	0.435413649	0.532113921	0.11867237	0.268526875	0.435413649	0.532113921	0.11867237	0.268526875
ATAD3C	1	1449689...	17	ENSG000	20470	3976	protein_c	219293	ATPase fa	3	-	-	GO:00055	-	-	1	1	0.079161	-1.1197	0.049064947	0.051832395	0.019912772	0.009163651	0.049064947	0.051832395	0.019912772	0.009163651
ATAD3B	1	1471769...	25	ENSG000	26080	7682	protein_c	83858	ATPase fa	5	-	GO:00057	GO:00055	-	-	2.85E-11	1	-0.11426	-0.16154	0.6868253	0.634526973	0.337685015	0.301913919	0.6868253	0.634526973	0.337685015	0.301913919
ATAD3A	1	1512151...	24	ENSG000	22537	3069	protein_c	55210	ATPase fa	7	GO:00160	GO:00057	GO:00055	-	-	1	1	-0.14881	-0.02272	0.360381904	0.32506328	0.149925832	0.147583269	0.360381904	0.32506328	0.149925832	0.147583269
TMEM24	1	compleme	9	ENSG000	5847	2009	protein_c	339453	transmem	3	-	GO:00300	-	-	-	1	1.31E-95	-0.29863	0.383807	1.132380104	0.920652409	0.091238469	0.119046142	1.132380104	0.920652409	0.091238469	0.119046142
SSU72	1	compleme	11	ENSG000	33191	6919	protein_c	29101	SSU72 ho	4	GO:00063	GO:00056	GO:00055	hsa03015	-	1	1	-0.00626	-0.46757	0.586224399	0.583687859	0.739809049	0.535016056	0.586224399	0.583687859	0.739809049	0.535016056
FNDC10	1	compleme	1	ENSG000	2124	2124	protein_c	643988	fibronecti	1	-	GO:00160	-	-	-	1	1	0.016865	-0.24113	0.054285851	0.054924161	0.133249698	0.11274034	0.054285851	0.054924161	0.133249698	0.11274034
MB2	1	1615415...	84	ENSG000	15196	8767	protein_c	142678	mindbomb	32	GO:00072	GO:00058	GO:00055	hsa05417	-	1.53E-32	9.11E-27	0.012108	0.070084	1.499254706	1.511890252	0.613570865	0.644112925	1.499254706	1.511890252	0.613570865	0.644112925
MMP23B	1	1632095...	27	ENSG000	3169	2273	protein_c	8510	matrix me	10	-	-	-	-	-	1	1	-0.05334	1.146583	0.175106147	0.168750166	0.014245262	0.031537437	0.175106147	0.168750166	0.014245262	0.031537437
CDK11B	1	compleme	37	ENSG000	23786	3488	protein_c	984	cyclin der	9	GO:00517	GO:00057	GO:00055	-	-	1	1	0.324923	0.1581	0.337265566	0.422458048	0.221823553	0.247514343	0.337265566	0.422458048	0.221823553	0.247514343
SLC35E2	1	compleme	18	ENSG000	33200	8395	protein_c	728661	solute car	5	-	GO:00160	-	-	-	1	1	0	0	0.000001	0.000001	0.000001	0.000001	0	0	0	0
CDK11A	1	compleme	63	ENSG000	21979	5085	protein_c	984	cyclin der	23	GO:00064	GO:00056	GO:00046	-	-	1	1	-0.15415	-0.47238	0.100989668	0.090755809	0.145997056	0.105230833	0.100989668	0.090755809	0.145997056	0.105230833
SLC35E2A	1	compleme	15	ENSG000	21162	7713	protein_c	9906	solute car	6	-	GO:00160	-	-	-	1	1	0	0	0.000001	0.000001	0.000001	0.000001	0	0	0	0
NADK	1	compleme	46	ENSG000	29226	5570	protein_c	65220	NAD kina	17	GO:00460	GO:00058	GO:00468	hsa00760	-	1	0.000799	-0.14356	0.149267	0.684798247	0.756447046	0.887582243	0.98433412	0.684798247	0.756447046	0.887582243	0.98433412
GNB1	1	compleme	26	ENSG000	105833	6362	protein_c	2782	G protein	9	GO:00072	GO:00973	GO:00048	hsa04014	-	1	2.46E-12	-0.19404	0.028944	3.372364332	2.947957596	2.844790329	2.902440855	3.372364332	2.947957596	2.844790329	2.902440855
CALML6	1	1915108...	11	ENSG000	2189	2040	protein_c	163688	calmoduli	4	-	GO:00057	GO:00055	hsa04014	D04789 L	1	1	-0.14849	-0.25798	0.050028221	0.045135134	0.025760725	0.021542663	0.050028221	0.045135134	0.025760725	0.021542663
TMEM52	1	compleme	17	ENSG000	1690	1600	protein_c	339456	transmem	6	-	GO:00160	-	-	-	1	1	0.949677	0.728437	0.008149146	0.015739589	0.003477414	0.00576153	0.008149146	0.015739589	0.003477414	0.00576153
CFAP74	1	compleme	56	ENSG000	81887	10422	protein_c	85452	cilia and f	10	-	-	-	-	-	1	1	0.038437	-0.52842	0.019072989	0.019587968	0.149767201	0.103835556	0.019072989	0.019587968	0.149767201	0.103835556
GABRD	1	2019329...	48	ENSG000	11430	4956	protein_c	2563	gamma-ar	18	GO:00068	GO:00300	GO:00052	hsa04080	D00058 g	5.53E-95	1.29E-40	0.088922	-1.02328	0.069298449	0.073704116	0.064271292	0.031621267	0.069298449	0.073704116	0.064271292	0.031621267
PRKCZ	1	2050411...	65	ENSG000	134985	7695	protein_c	5590	protein kin	26	GO:00602	GO:00484	GO:00046	hsa04015	-	1E-179	7.08E-79	-0.03946	0.475326	0.263386349	0.256280154	0.159678382	0.221990072	0.263386349	0.256280154	0.159678382	0.221990072
FAAP20	1	compleme	40	ENSG000	28260	10916	protein_c	199990	FA core c	15	-	-	-	-	-	2.25E-53	1	-0.09244	0.080612	1.43543603	1.346345072	0.693603	0.733461741	1.43543603	1.346345072	0.693603	0.733461741
SKI	1	2228319...	13	ENSG000	81895	6550	protein_c	6497	SKI proto	4	GO:00303	GO:00056	GO:00463	-	-	1	1.38E-47	-0.07685	-0.07462	2.183982274	2.070687397	3.800423881	3.608863507	2.183982274	2.070687397	3.800423881	3.608863507
MORN1	1	compleme	31	ENSG000	70455	9098	protein_c	79906	MORN re	10	-	-	-	-	-	1	1	0	0	0.000001	0.000001	0.000001	0.000001	0	0	0	0
RER1	1	2391775...	25	ENSG000	13668	5689	protein_c	11079	retention f	9	GO:00713	GO:00301	GO:00331	-	-	1	1	-0.0184	-0.0812	1.342449713	1.325437593	1.344587566	1.270997668	1.342449713	1.325437593	1.344587566	1.270997668
PEX10	1	compleme	28	ENSG000	9834	3789	protein_c	5192	peroxisom	9	GO:00165	GO:00057	GO:00055	hsa04146	-	1	1	-0.07453	0.252624	0.440949387	0.418749634	0.213542987	0.254409202	0.440949387	0.418749634	0.213542987	0.254409202
PLCH2	1	2425980...	41	ENSG000	79553	10809	protein_c	9651	phospholi	8	GO:00436	GO:00057	GO:00036	hsa00562	-	6E-49	5.76E-44	-0.09876	4.147616	0.204125741	0.190619257	0.00479048	0.084905479	0.204125741	0.190619257	0.00479048	0.084905479

◆ Pathway enrichment Analysis

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KEGG: Kyoto Encyclopedia of Genes and Genomes

KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular level information, especially large-scale molecular datasets generated by high-throughput technologies.

Category	PathwayID	Pathway description	Number of DEGs with pathway annotation (811)	Number of DEGs with pathway annotation % (811)	Up regulated	Down regulated	Number of All genes with pathway annotation (7815)	Number of All genes with pathway annotation % (7815)	Rich factor	P-value	Q-value	Bonferroni	Benjamini	FDR(%)	Gene(Up)	Gene(Down)
KEGG PATHWAY	hsa04740	Olfactory transduction	11	1.36%	6	5	439	5.62%	2.51%	7.00E-10	2.13E-07	2.13E-07	2.13E-07	2.13E-05	CAMK2B(8),GRIK1(8),GRIK2(8),GRIK3(8),GRIK4(8),GRIK5(8),GRIK6(8),GRIK7(8),GRIK8(8),GRIK9(8),GRIK10(8),GRIK11(8),GRIK12(8),GRIK13(8),GRIK14(8),GRIK15(8),GRIK16(8),GRIK17(8),GRIK18(8),GRIK19(8),GRIK20(8),GRIK21(8),GRIK22(8),GRIK23(8),GRIK24(8),GRIK25(8),GRIK26(8),GRIK27(8),GRIK28(8),GRIK29(8),GRIK30(8),GRIK31(8),GRIK32(8),GRIK33(8),GRIK34(8),GRIK35(8),GRIK36(8),GRIK37(8),GRIK38(8),GRIK39(8),GRIK40(8),GRIK41(8),GRIK42(8),GRIK43(8),GRIK44(8),GRIK45(8),GRIK46(8),GRIK47(8),GRIK48(8),GRIK49(8),GRIK50(8),GRIK51(8),GRIK52(8),GRIK53(8),GRIK54(8),GRIK55(8),GRIK56(8),GRIK57(8),GRIK58(8),GRIK59(8),GRIK60(8),GRIK61(8),GRIK62(8),GRIK63(8),GRIK64(8),GRIK65(8),GRIK66(8),GRIK67(8),GRIK68(8),GRIK69(8),GRIK70(8),GRIK71(8),GRIK72(8),GRIK73(8),GRIK74(8),GRIK75(8),GRIK76(8),GRIK77(8),GRIK78(8),GRIK79(8),GRIK80(8),GRIK81(8),GRIK82(8),GRIK83(8),GRIK84(8),GRIK85(8),GRIK86(8),GRIK87(8),GRIK88(8),GRIK89(8),GRIK90(8),GRIK91(8),GRIK92(8),GRIK93(8),GRIK94(8),GRIK95(8),GRIK96(8),GRIK97(8),GRIK98(8),GRIK99(8),GRIK100(8),GRIK101(8),GRIK102(8),GRIK103(8),GRIK104(8),GRIK105(8),GRIK106(8),GRIK107(8),GRIK108(8),GRIK109(8),GRIK110(8),GRIK111(8),GRIK112(8),GRIK113(8),GRIK114(8),GRIK115(8),GRIK116(8),GRIK117(8),GRIK118(8),GRIK119(8),GRIK120(8),GRIK121(8),GRIK122(8),GRIK123(8),GRIK124(8),GRIK125(8),GRIK126(8),GRIK127(8),GRIK128(8),GRIK129(8),GRIK130(8),GRIK131(8),GRIK132(8),GRIK133(8),GRIK134(8),GRIK135(8),GRIK136(8),GRIK137(8),GRIK138(8),GRIK139(8),GRIK140(8),GRIK141(8),GRIK142(8),GRIK143(8),GRIK144(8),GRIK145(8),GRIK146(8),GRIK147(8),GRIK148(8),GRIK149(8),GRIK150(8),GRIK151(8),GRIK152(8),GRIK153(8),GRIK154(8),GRIK155(8),GRIK156(8),GRIK157(8),GRIK158(8),GRIK159(8),GRIK160(8),GRIK161(8),GRIK162(8),GRIK163(8),GRIK164(8),GRIK165(8),GRIK166(8),GRIK167(8),GRIK168(8),GRIK169(8),GRIK170(8),GRIK171(8),GRIK172(8),GRIK173(8),GRIK174(8),GRIK175(8),GRIK176(8),GRIK177(8),GRIK178(8),GRIK179(8),GRIK180(8),GRIK181(8),GRIK182(8),GRIK183(8),GRIK184(8),GRIK185(8),GRIK186(8),GRIK187(8),GRIK188(8),GRIK189(8),GRIK190(8),GRIK191(8),GRIK192(8),GRIK193(8),GRIK194(8),GRIK195(8),GRIK196(8),GRIK197(8),GRIK198(8),GRIK199(8),GRIK200(8),GRIK201(8),GRIK202(8),GRIK203(8),GRIK204(8),GRIK205(8),GRIK206(8),GRIK207(8),GRIK208(8),GRIK209(8),GRIK210(8),GRIK211(8),GRIK212(8),GRIK213(8),GRIK214(8),GRIK215(8),GRIK216(8),GRIK217(8),GRIK218(8),GRIK219(8),GRIK220(8),GRIK221(8),GRIK222(8),GRIK223(8),GRIK224(8),GRIK225(8),GRIK226(8),GRIK227(8),GRIK228(8),GRIK229(8),GRIK230(8),GRIK231(8),GRIK232(8),GRIK233(8),GRIK234(8),GRIK235(8),GRIK236(8),GRIK237(8),GRIK238(8),GRIK239(8),GRIK240(8),GRIK241(8),GRIK242(8),GRIK243(8),GRIK244(8),GRIK245(8),GRIK246(8),GRIK247(8),GRIK248(8),GRIK249(8),GRIK250(8),GRIK251(8),GRIK252(8),GRIK253(8),GRIK254(8),GRIK255(8),GRIK256(8),GRIK257(8),GRIK258(8),GRIK259(8),GRIK260(8),GRIK261(8),GRIK262(8),GRIK263(8),GRIK264(8),GRIK265(8),GRIK266(8),GRIK267(8),GRIK268(8),GRIK269(8),GRIK270(8),GRIK271(8),GRIK272(8),GRIK273(8),GRIK274(8),GRIK275(8),GRIK276(8),GRIK277(8),GRIK278(8),GRIK279(8),GRIK280(8),GRIK281(8),GRIK282(8),GRIK283(8),GRIK284(8),GRIK285(8),GRIK286(8),GRIK287(8),GRIK288(8),GRIK289(8),GRIK290(8),GRIK291(8),GRIK292(8),GRIK293(8),GRIK294(8),GRIK295(8),GRIK296(8),GRIK297(8),GRIK298(8),GRIK299(8),GRIK300(8),GRIK301(8),GRIK302(8),GRIK303(8),GRIK304(8),GRIK305(8),GRIK306(8),GRIK307(8),GRIK308(8),GRIK309(8),GRIK310(8),GRIK311(8),GRIK312(8),GRIK313(8),GRIK314(8),GRIK315(8),GRIK316(8),GRIK317(8),GRIK318(8),GRIK319(8),GRIK320(8),GRIK321(8),GRIK322(8),GRIK323(8),GRIK324(8),GRIK325(8),GRIK326(8),GRIK327(8),GRIK328(8),GRIK329(8),GRIK330(8),GRIK331(8),GRIK332(8),GRIK333(8),GRIK334(8),GRIK335(8),GRIK336(8),GRIK337(8),GRIK338(8),GRIK339(8),GRIK340(8),GRIK341(8),GRIK342(8),GRIK343(8),GRIK344(8),GRIK345(8),GRIK346(8),GRIK347(8),GRIK348(8),GRIK349(8),GRIK350(8),GRIK351(8),GRIK352(8),GRIK353(8),GRIK354(8),GRIK355(8),GRIK356(8),GRIK357(8),GRIK358(8),GRIK359(8),GRIK360(8),GRIK361(8),GRIK362(8),GRIK363(8),GRIK364(8),GRIK365(8),GRIK366(8),GRIK367(8),GRIK368(8),GRIK369(8),GRIK370(8),GRIK371(8),GRIK372(8),GRIK373(8),GRIK374(8),GRIK375(8),GRIK376	

Enriched Pathway - Scatter Plot

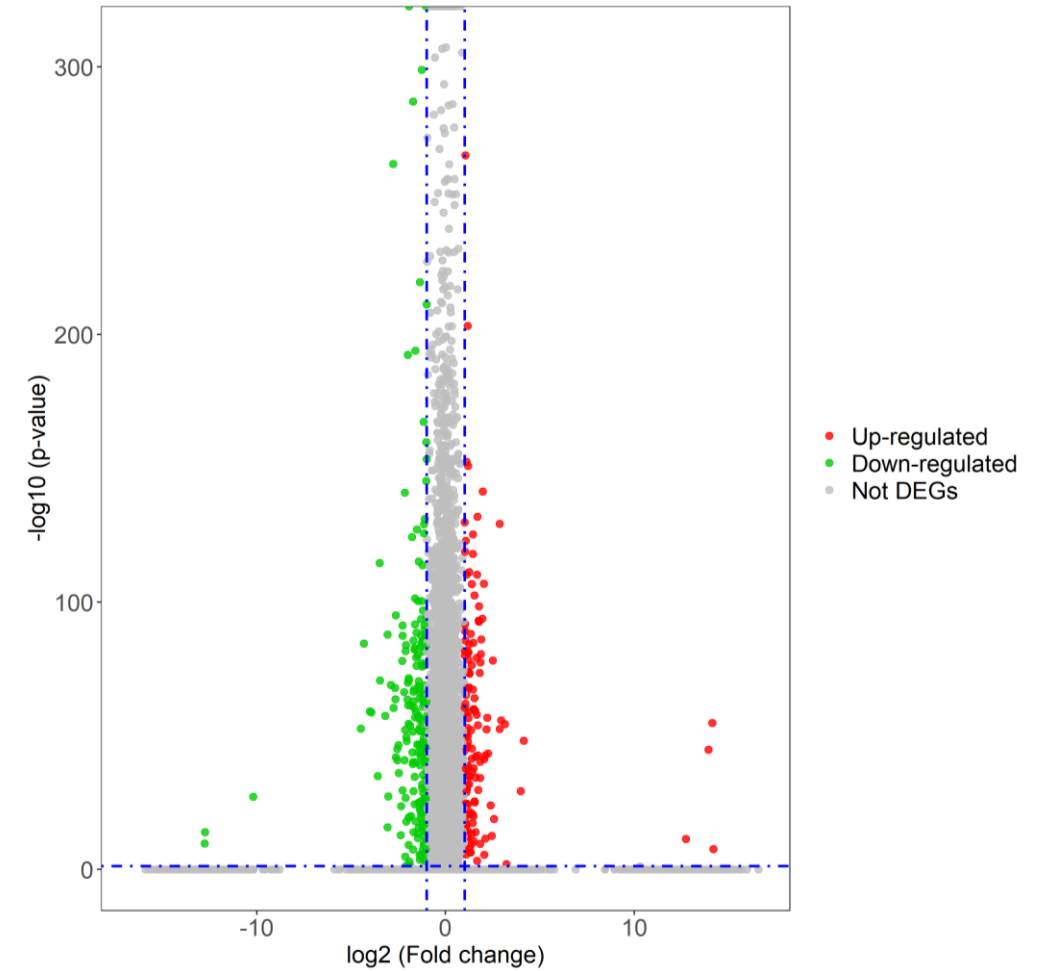
Statistics of Pathway Enrichment

Rich factor

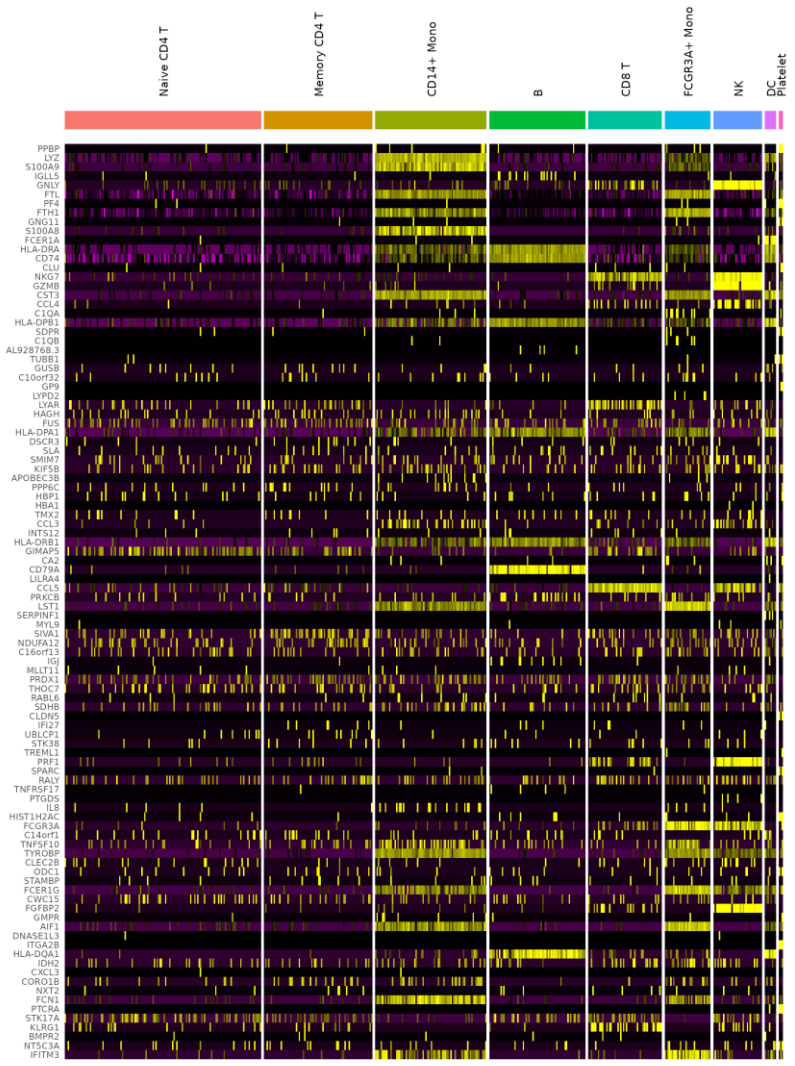
pvalue

Gene number

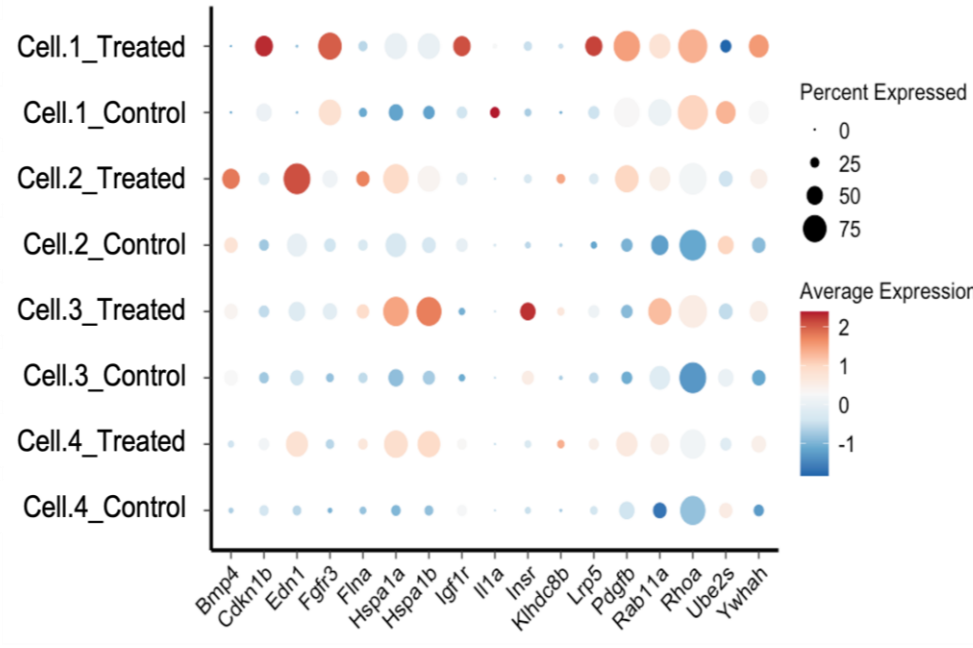
Pathway	Rich factor (approx.)	pvalue (approx.)	Gene number (approx.)
Ubiquitin mediated proteolysis	0.05	0.0025	20
Thermogenesis	0.05	0.0025	20
Spliceosome	0.05	0.0025	20
Salmonella infection	0.05	0.0025	20
Ribosome biogenesis in eukaryotes	0.05	0.0025	20
Ribosome	0.05	0.0025	20
Protein processing in endoplasmic reticulum	0.05	0.0025	20
Prion disease	0.05	0.0025	20
Olfactory transduction	0.05	0.0025	20
Nucleocytoplasmic transport	0.05	0.0025	20
Non-alcoholic fatty liver disease	0.05	0.0025	20
Neuroactive ligand-receptor interaction	0.20	0.0015	60
mRNA surveillance pathway	0.05	0.0025	20
Insulin secretion	0.25	0.0015	40
Gastric acid secretion	0.25	0.0015	40
Cholinergic synapse	0.25	0.0015	40
Cell cycle	0.05	0.0025	20
Calcium signaling pathway	0.25	0.0015	40
Basal cell carcinoma	0.30	0.0015	40
Autophagy - animal	0.05	0.0025	20



Heatmap



Dotplot of features Analysis

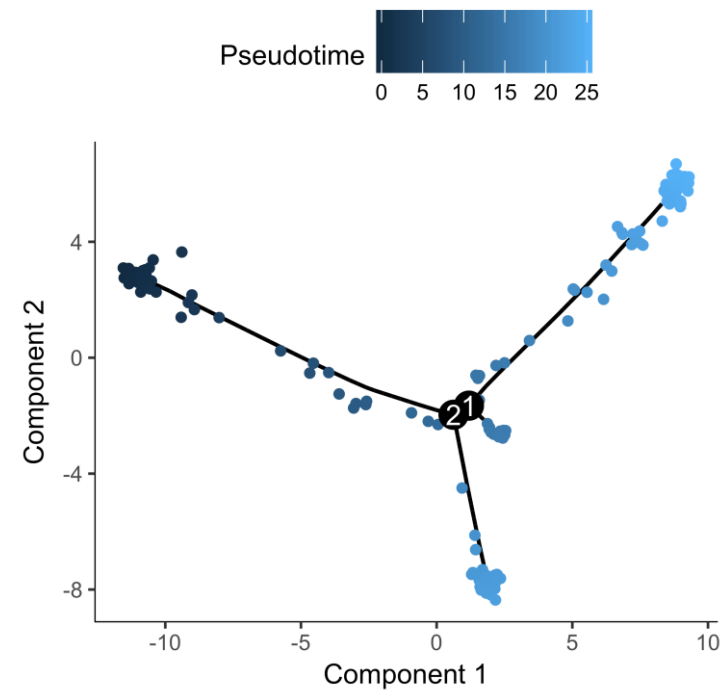
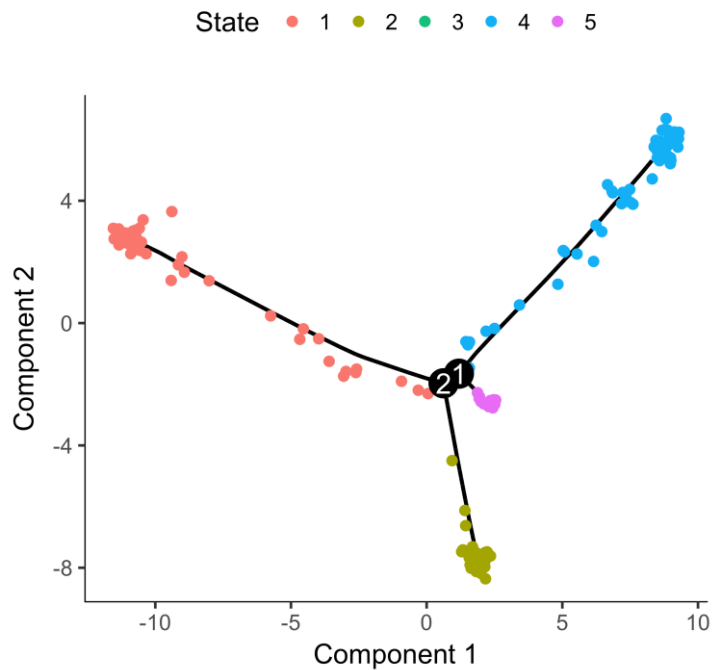
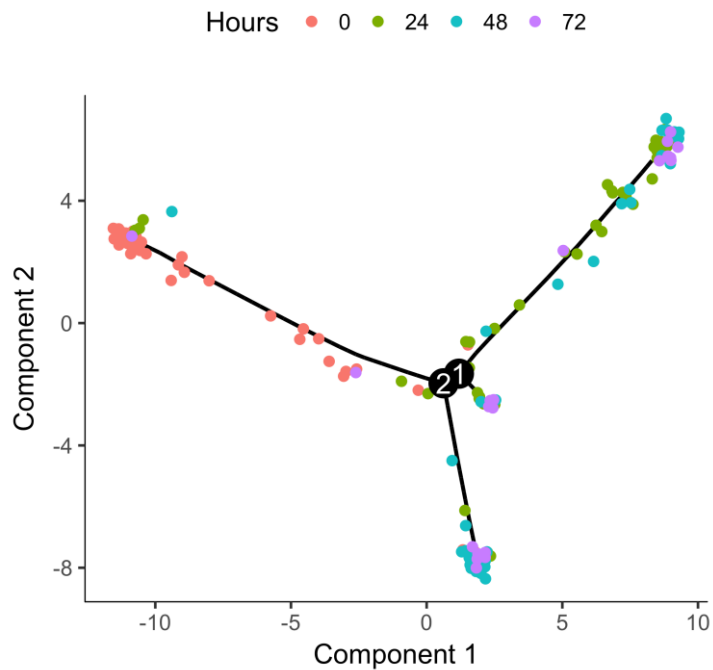


Seurat

◆ Trajectory Analysis (Pseudotime)

- 使用Monocle2 package進行Trajectory Analysis

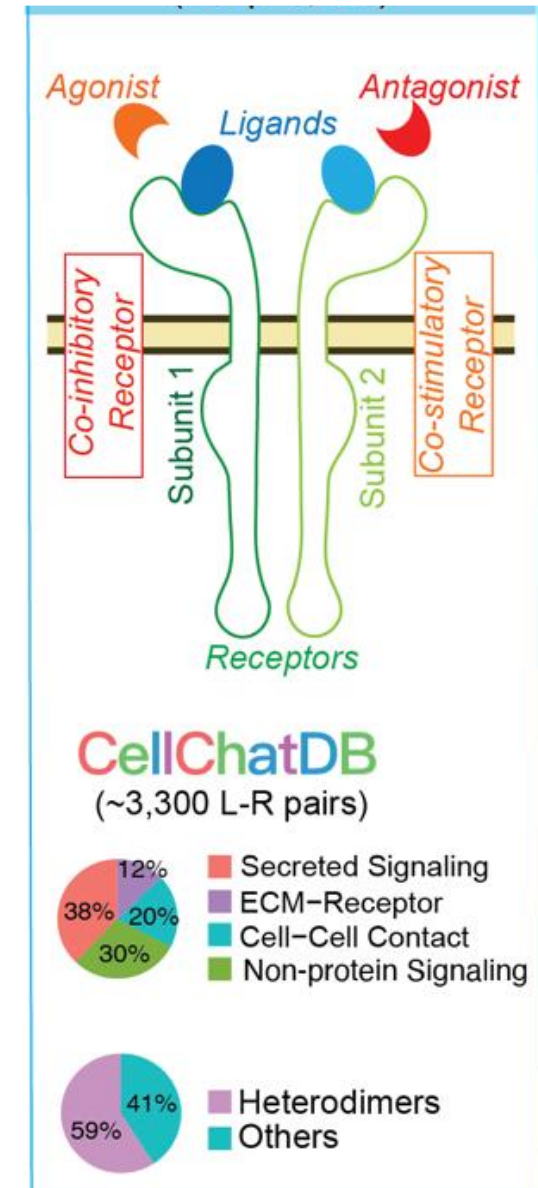
<https://cole-trapnell-lab.github.io/monocle-release/docs/>



◆ CellChat

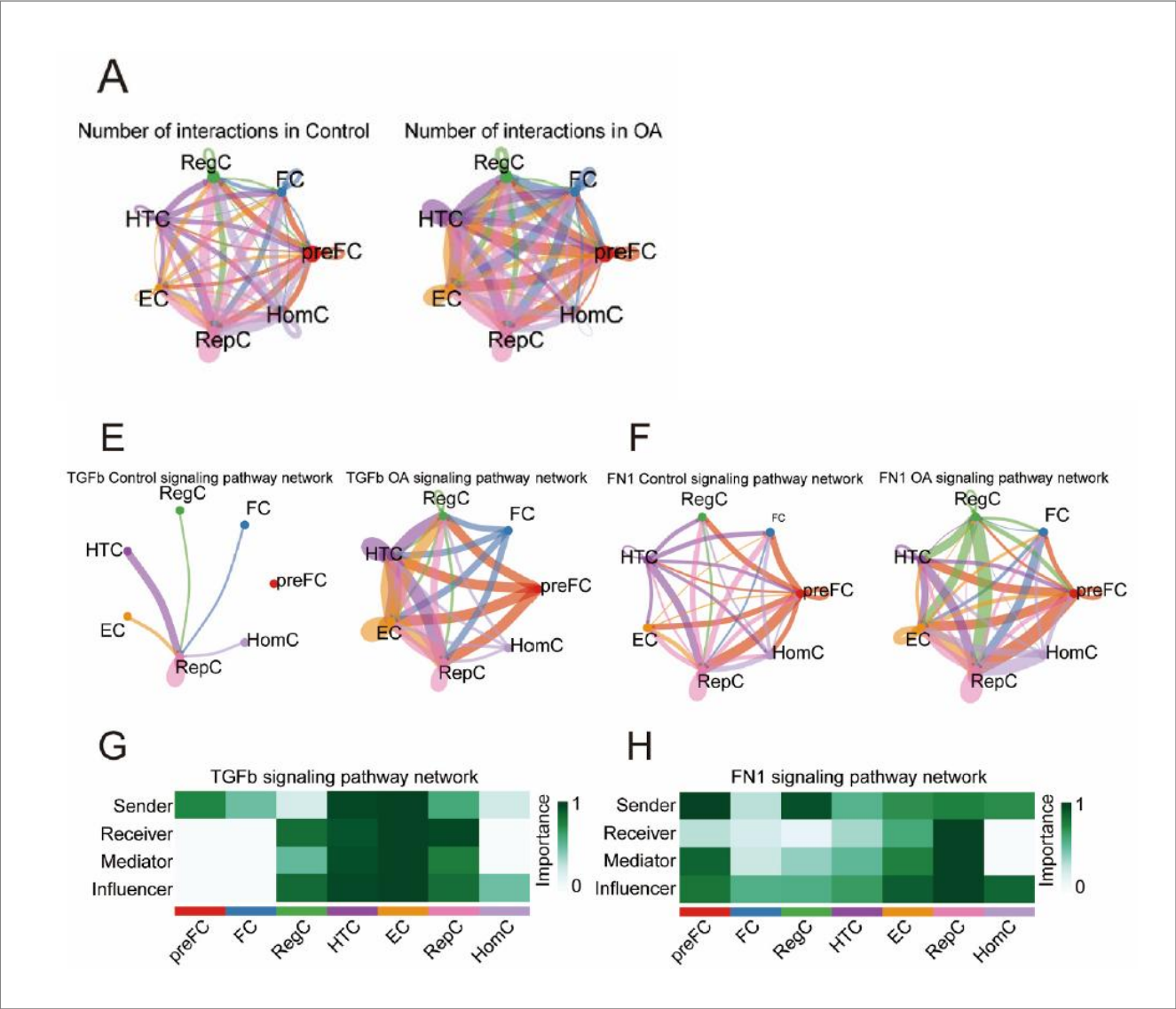


- **systematic analysis of cell-cell communication**
- CellChat quantifies the signaling communication probability between two cell groups using a simplified mass action-based model, which incorporates the core interaction between ligands and receptors with multi-subunit structure along with modulation by cofactors.
- **CellChatDB database:**
 1. incorporates multi-subunit structure of ligand-receptor complexes
 2. accounts for soluble and membrane-bound stimulatory and inhibitory cofactors such as agonists, antagonists and co-receptors
- **CellChatDB database:**
 1. signaling interactions
 2. extracellular matrix (ECM)-receptor interactions
 3. cell-cell contact interactions as well as interactions involving non-proteins acting as ligands



Nat Protoc 20, 180–219 (2025).

<https://github.com/jinworks/CellChat>





NGS & 第三代定序

高效率的定序技術，再結合生物資訊的分析技術，已廣泛地應用在各個不同的學術領域研究基因體、轉錄體及表觀基因體等。

[NGS introduction](#)

[Genome sequencing](#)

[Metagenome sequencing](#)

[Epigenome sequencing](#)

[RNA sequencing](#)

[Ribo-seq](#)

[Single-cell RNA-seq](#)

1984 年人類基因體計畫 (Human Genome Project, HGP) 以第一代定序方法(Sanger Sequencing)，用了超過10年的時間且花費30 億美元完成整個人類基因體30億個鹼基對的定序。

隨著定序技術的演進與發展，2007年次世代定序 (Next Generation Sequencing, NGS) 的技術問世，大幅降低定序所需的成本和時間。次世代定序又稱為第二代定序或基因高通量分析(High-Throughput Analysis)，此高效率的定序技術至今單次輸出量已可高達540-600 Gb (Illumina system)，再結合上生物資訊的分析技術，次世代定序已被廣泛應用在各個不同的學術領域研究基因體、轉錄體及表觀基因體等。

第三代定序(Third Generation Sequencing)，又可稱為「長讀取定序技術」(Long-Read Sequencing)，目前平均的序列讀取長度可大於1 萬個鹼基(10 kb)，能直接定序原始DNA 樣本，省去次世代定序技術需要的PCR增幅過程，可避開 PCR 擴增的錯誤率及偏好性問題，也能分析DNA 甲基化修飾。較廣為人知的技術有Pacific BioSciences (PacBio) 公司2010 年推出的「單分子即時定序」(Singlemolecule Real-Time Sequencing, SMRT)，以及Oxford Nanopore Technologies (ONT) 公司於2012 年底推出的「奈米孔定序」(Nanopore-Based Platform)。第三代定序系統的解序片段長，除了上述的基因體、轉錄體、表觀基因體研究，於生物醫學和生物技術等領域具有更廣泛的應用。並且，第三代定序相對於第一代定序、第二代定序，在改善速度、降低成本和提高數據質量等方面也有顯著的進展，為研究人員的需求提供了更多的選擇。源資具備Nanopore平台的Flongle, MinION, GridION, 以及2022年底最新推出的PromethION 2 Solo。



官網 QR code



官方LINE QR code



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